Task 3a. Data base design in Data Warehousing Process

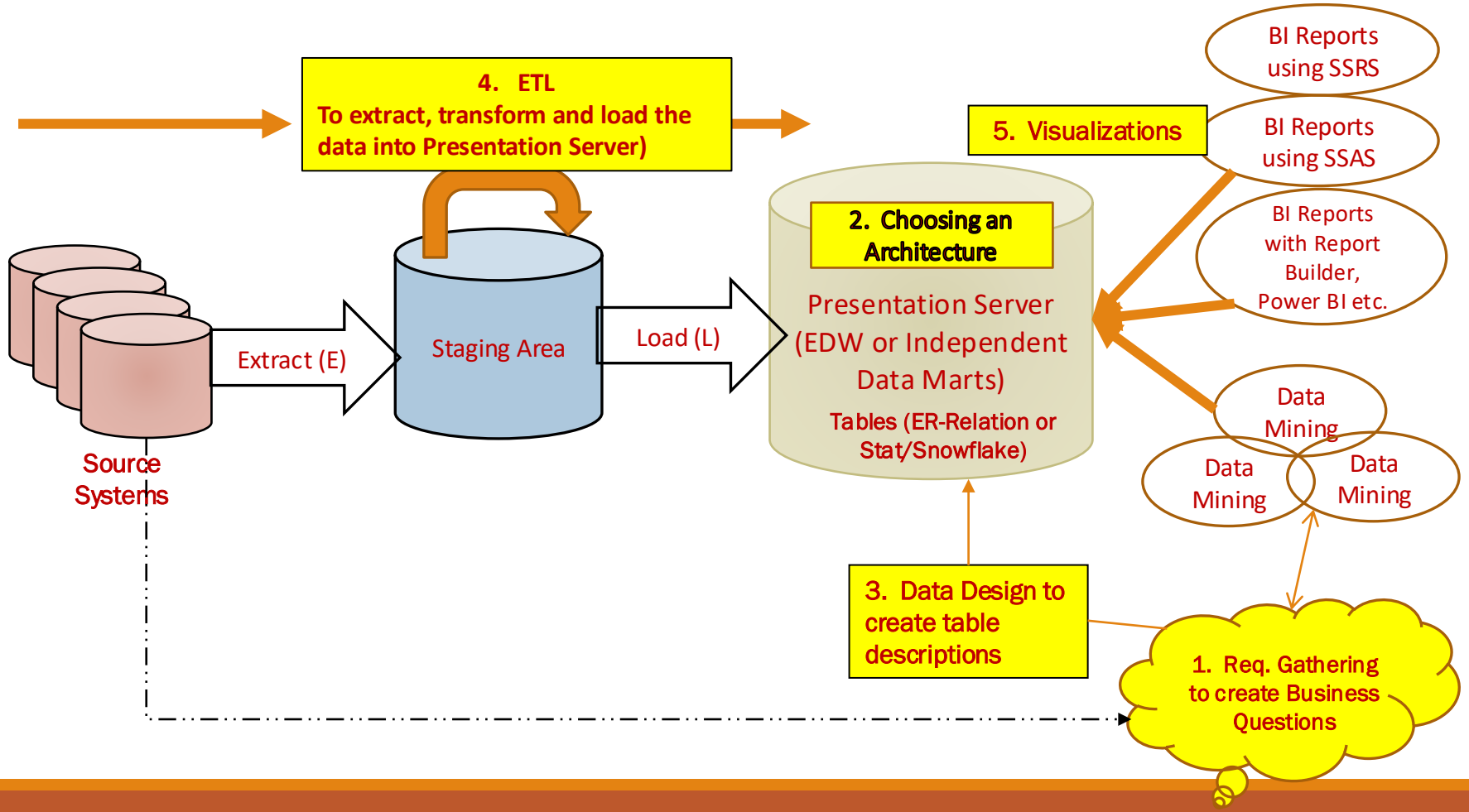
ARUN SEN

DEPARTMENT OF INFORMATION & OPERATIONS MGMT.

MAYS BUSINESS SCHOOL

TEXAS A&M UNIVERSITY

Following the Data Warehousing Process



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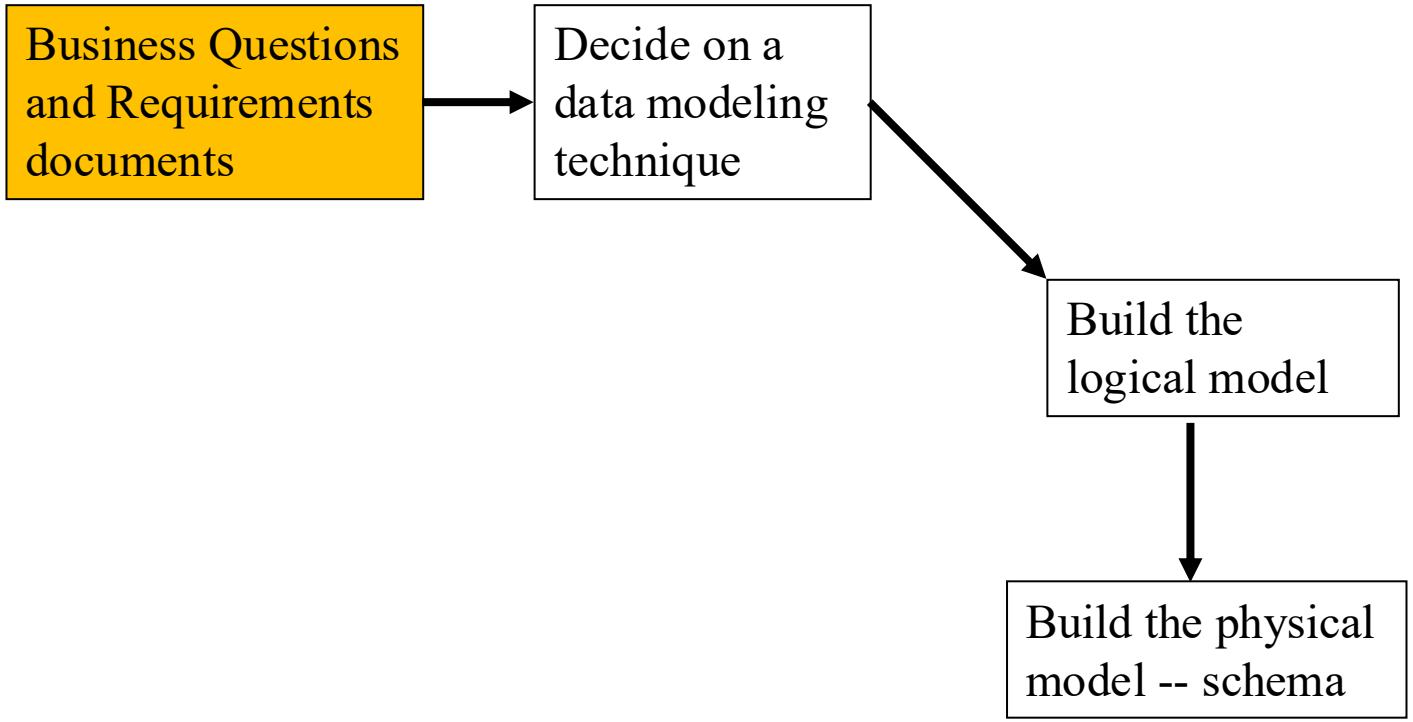
Data Design for Data Warehousing

A. Data Design Process used by Data Architect



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Data Design for Data Warehousing

B. Terms to be used in Data Modeling --

- **Conceptual Model** -- world model -- ERD
- **Logical Model** -- model used by a DBMS - RDM
- **Physical Model** – adding physical characteristics like indexing style, etc.
- **Implementation model** -- use of Oracle or SQL Server style

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Data Design for Data Warehousing

C. Two Choices in Data Design --

I. ERD (conceptual) and Relational Schema (logical)

II. Dimensional model (conceptual+logical) and Star/Snowflake Schema (logical)

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I. ERD (conceptual) & Relational Schema (logical) – Targeted for EDW

I. ERD and Relational Schema

Traditional data base design

- ***Step 1:*** Build the ERD for the enterprise (either directly or through conformable data marts) - - "conceptual model"
- ***Step 2:*** Translate it to a set of relations using translation rules
- ***Step 3:*** Evaluate the relations (called a process of Normalization with an objective to have all relations in the "3NF model"
- ***Step 4:*** **Denormalize as needed**

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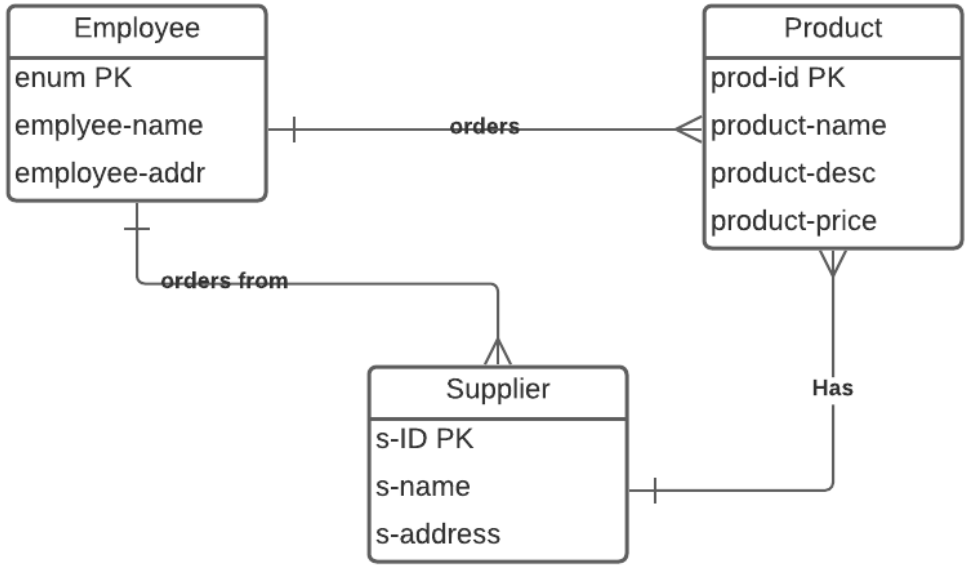
I. ERD and Relational Schema

Step 1



To understand database design, watch this video: https://www.youtube.com/watch?v=I_rxqSJAj6U

For Lucid chart https://lucid.app/lucidchart/6f67e4b8-b2d4-40c5-b61e-a8b426c2f487/edit?beaconFlowId=81B2705DE817D131&page=0_0#?folder_id=home&browser=icon

Download Visio from https://azureforeducation.microsoft.com/devtools

I. ERD and Relational Schema

Step 2. Translated Relations are as follows:

1. Employee (<u>enum</u>, employee-name, employee-addr)
2. Supplier (<u>s-ID</u>, s-name, s-address, <u>enum</u>)
3. Product (<u>prod-id</u>, product-name, product-desc, product-price, <u>enum</u>, <u>s-ID</u>)

Translation Rule 1
[a] Each entity type in an ERD (that is not "weak") is transformed to a relation.
[b] The PK of the entity becomes the PK of the relation
[c] The non-key attributes of the entity become the non-key attributes of the relation.

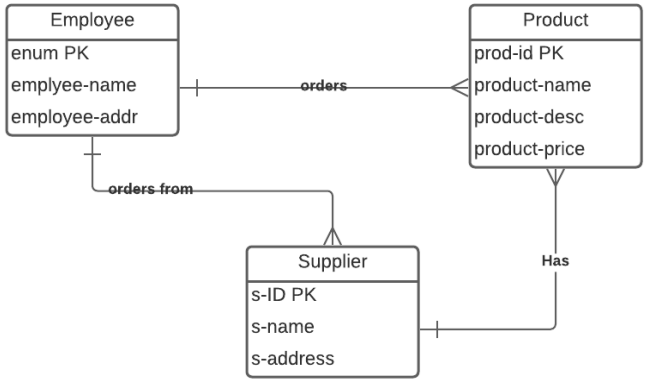
Translation Rule 2
[a]For each weak entity type generate a new relation.
[b] The PK is the composite key created by concatenation of its own key and primary key of the supporting entity.
[c] The non-key attributes of the entity become the non-key attributes of the relation.

Translation Rule 3
For each n-ary relationship, generate a corresponding relation. The key is the "composite key" consisting of primary key for each of the entities in the relationship.

Translation Rule 4
For each binary relationship WITH attributes, generate a new relation. The key of this relation is the "composite key" consisting of primary key for each of the entities in the relationship.

Translation Rule 5
For each binary relationship WITHOUT attributes, generate a new relation only if the relationship is (many, many). The key of this relation is the composite key consisting of primary key for each of the two entities in the relationship.

Translation Rule 6
For each binary relationship WITHOUT attributes, if the relationship is (1,many) or (1,1), place a foreign key in the relation that represents the entity on the many-side of the relationship. The foreign key is the primary-key of the entity on one-side of the relationship. For (1,1), the foreign key can be placed in either relation.



Step 2

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I. ERD and Relational Schema

Step 3

First normal form (1NF) is a property of a relation in a relational database. **A relation is in first normal form if and only if the domain of each attribute contains only atomic (indivisible) values, and the value of each attribute contains only a single value from that domain.**

A relation is in the second normal (2NF) form if it fulfills the following two requirements:

1. **It is in first normal form.**
2. **It does not have any non-prime attribute that is functionally dependent on any proper subset of any candidate key of the relation. A non-prime attribute of a relation is an attribute that is not a part of any candidate key of the relation.**

A database relation is said to meet third normal form (3NF) standards if all the attributes (e.g. database columns) are functionally dependent on solely the primary key. **Codd defined this as a relation in second normal form where all non-prime attributes depend only on the candidate keys and do not have a transitive dependency on another key.**

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I. ERD and Relational Schema

Step 3

- https://www.geeksforgeeks.org/normal-forms-in-dbms/ to understand normal forms

- Based up on Normal form definitions, we evaluate the relations with an objective to create Third Normal From relations as follows:
  1. Employee (enum, employee-name, employee-addr) -- 3rd Normal Form
  2. Supplier (s-ID, s-name, s-address) - - 3rd Normal form
  3. Order (s-ID, enum) -- 3rd Normal Form
  4. Product (prod-id, product-name, product-desc, product-price) -- 3rd Normal Form
  5. Supply (prod-id, enum, s-ID) -- 3rd Normal Form

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Denormalization of 3NF Relations

- Advantages:
  - -- improve performance
- Disadvantages
  - -- reduces data flexibility
  - -- introduces the potential for data anomalies
  - -- increases programming cost and complexity

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Denormalization of 3NF Relations

A. Recreating Repeating Column Groups

- *Why?*
  - A report is needed to compare and calculate percentages of an employee's sales for the **last** six months.

- *How is it done **now**?*
  - Have a 3NF Relation called Employee_Sales_Report (emp#, sales-date, sales-amount).
  - This is a history table that tends to grow very big over time.

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Denormalization of 3NF Relations

A. Recreating Repeating Column Groups

- ***Choice 1**: Denormalize inside the Logical Model*
  - Providing repeating groups with a meaningful name such January, February, etc. ... but SQL statement must be changed each time
  - Use of non-meaningful column names such as Month-1, Month-2, etc. ... leads to confusion as someone may not know what Month-1 means.
  - No matter how many columns we provide, there will be more needed ...

  - *Bad Choice*

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Denormalization of 3NF Relations

A. Recreating Repeating Column Groups

- ***Choice 2:***
  - Keep original in 3NF and add an additional temporary table created and refreshed every month.
  - Denormalization is outside the logical model ...
  - cost and complexity

- *Good Choice*

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Denormalization of 3NF Relations

B. Derived Data

- Derived data is defined as redundant stored data whose values can be determined or calculated from other column values.
- Example

Employee(emp#, lastname, store#)
Store (store#, storename)

- ***Choice***

We need to find employee count. We can find it by doing a total table scan (expensive!) of Employee table. Instead, we can have:

Store(store#, storename, employee-count)

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Denormalization of 3NF Relations

C. Pre-Joins

- Some DBMSs like Teradata RDBMS is capable of performing 64-table joins. But this is costly ...
- Try to avoid joins to minor relations by carrying their attributes in the major relations.
- Example:

Employee(emp#, lastname, store#) needs to be joined with Store(store#, storename)

- Example:

  Instead ...

  Employee(emp#, lastname, store#, storename)

  - Make sure to update properly. Also, keep the Store Relation to avoid anomalies.

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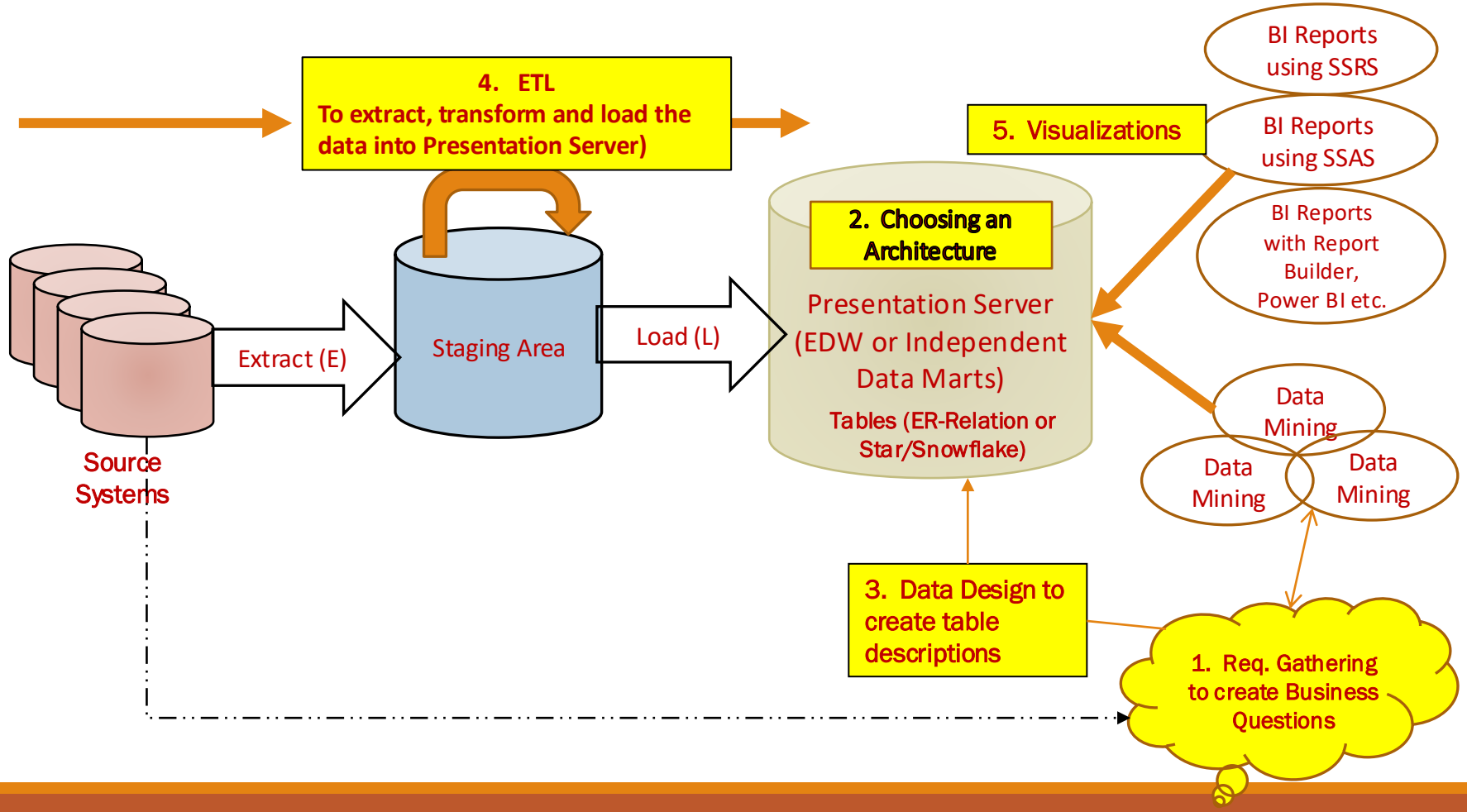
II. Dimensional model (conceptual+logical) and Star/Snowflake Schema (logical) – Targeted for Independent Data marts

Following the Data Warehousing Process



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II. Dimensional Modeling and Star/Snowflake Schema

1. History of Dimensional Modeling

- first used by General Mills and Dartmouth University in 1960's
- in 1970's it is used by Nielson Marketing
- in 1984 -- Kimball introduced it as a data modeling tool for conformable Data marts

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II. Dimensional Modeling and Star/Snowflake Schema

2. What is Dimensional Modeling?

- It is a logical design technique that seeks to present data in a standard format that is intuitive and allows for high-performance access
- Fundamental idea of dimensional modeling is that every type of business data can be represented as a Cube or Hypercube.
- More suitable for Kimball's strategy
- Not much use in creating an enterprise-wise star schema!!

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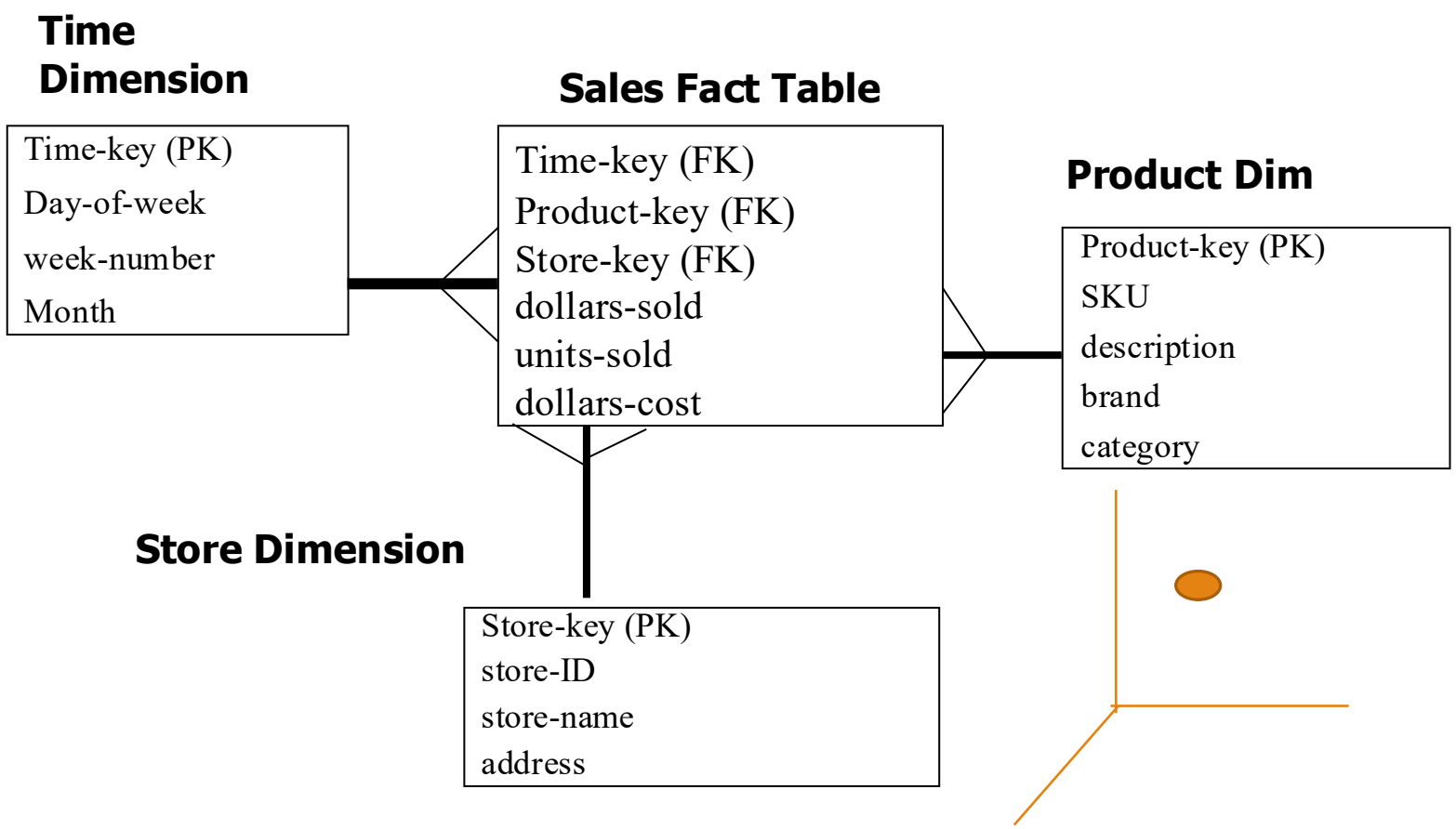
Star Schema



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II. Dimensional Modeling and Star/Snowflake Schema

3. Components of Dimensional Model

- It consists of two types of tables --
  - Fact table with attributes
  - Dimension table with attributes

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II. Dimensional Modeling and Star/Snowflake Schema

4. What is a Fact Table?

- It contains the observations of the market place
- Facts should be
  - Additive (e.g., sales data)
  - Non-additive (some facts could be non-additive such as account balance over time)
  - Semi-additive (some account balance could be additive along other dimensions)
  - Textual (keep them at minimum)

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4. What is a Fact Table?

- It is usually the m:n relationship of the ERD
- contains lot of FKs

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II. Dimensional Modeling and Star/Snowflake Schema

4. **What is a Fact Table?**
   - Characteristics of a fact table
     - it will contain concatenated key
     - data granularity is important for a fact table.
     - fully additive measures
     - semi-additive measures
     - table is deep and not wide (i.e., not too many attributes)
     - sparse data
     - Degenerate dimension

       Sometimes when we start to collect attributes from OLTP systems for dimension tables and fact tables, we may left out with attributes like Order# or Order-line#. These are neither facts nor strictly dimension attributes. Usually, we put them in the fact table are called “degenerate dimensions.” They are needed to relate orders to products

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II. Dimensional Modeling and Star/Snowflake Schema

5. Types of Fact Table?

- Different types of Fact Tables –
  - Single fact table
  - Multiple fact tables

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6. Single Fact Table

Sales Fact Table

Time-key (FK)
Product-key (FK)
Store-key (FK)
dollars-sold
units-sold
dollars-cost

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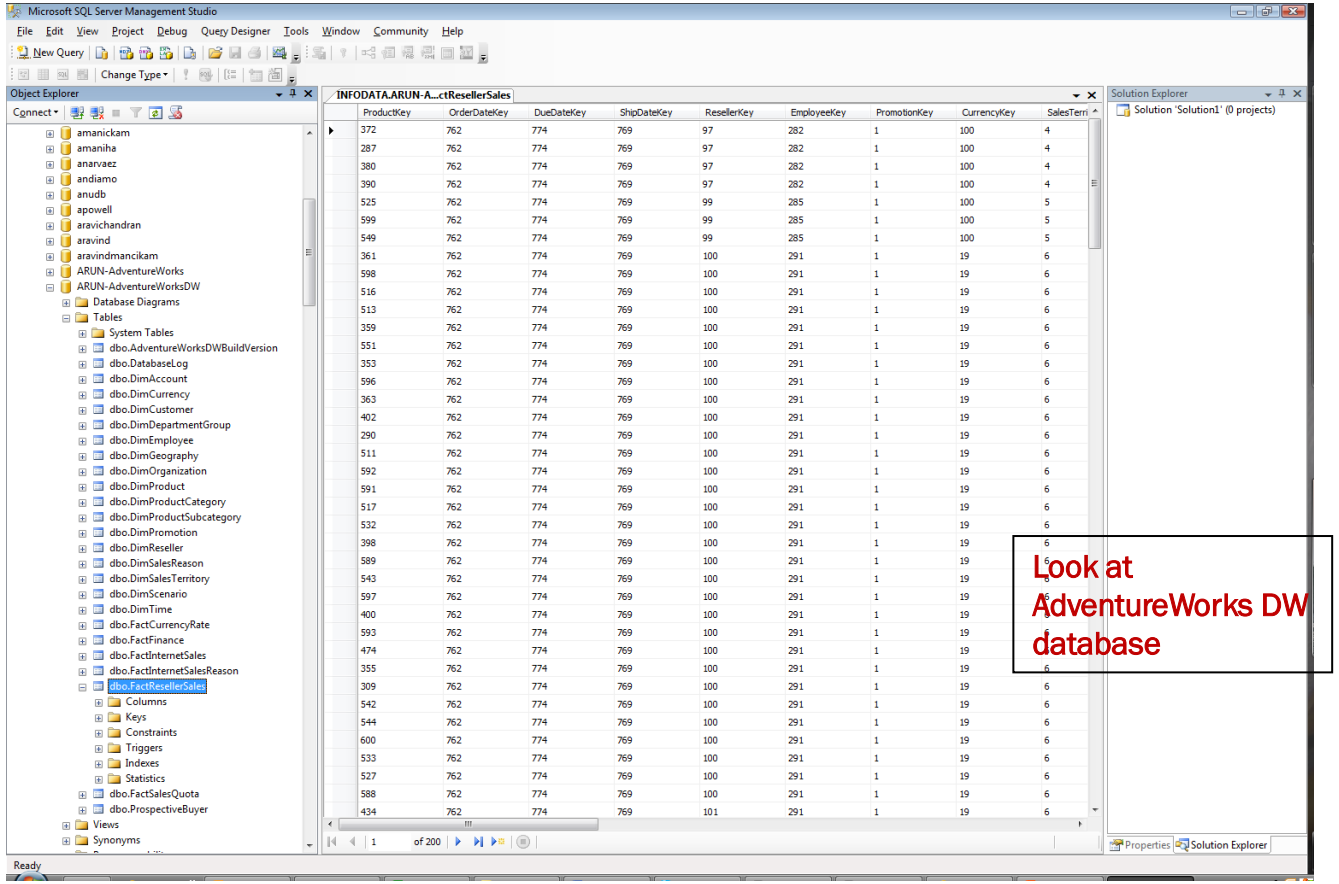
II. Dimensional Modeling and Star/Snowflake Schema

6. Single Fact Table



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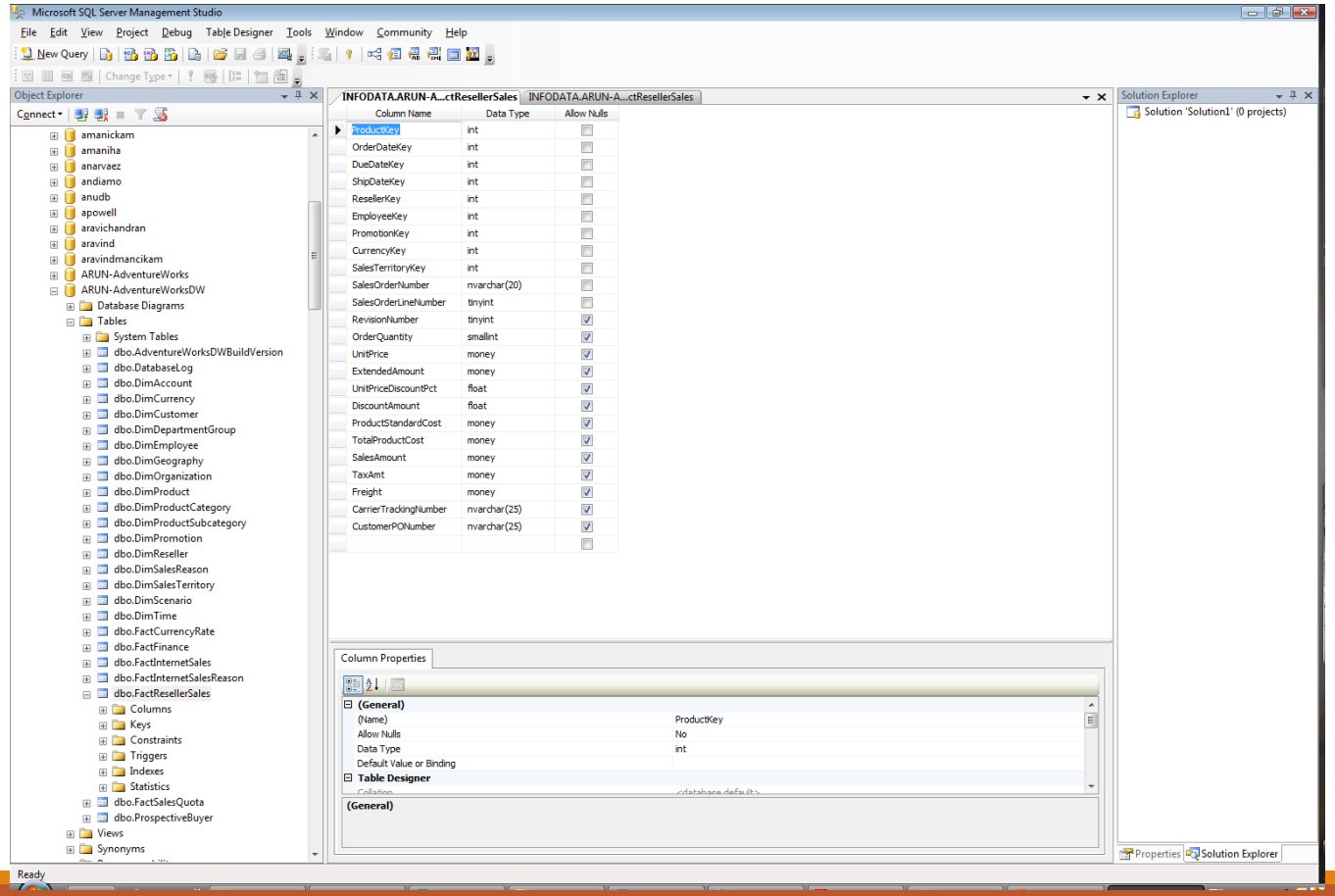
II. Dimensional Modeling and Star/Snowflake Schema

6. Single Fact Table



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II. Dimensional Modeling and Star/Snowflake Schema

6. Single Fact Table

- Factless Fact Table
  - A fact table can be factless if it records an **event** e.g., student attendance event
  - A fact table is factless if it records **coverage**. For example, if a fact table is used to store the promotion coverage only, it cannot have product data for those that did not sell in spite of promotions.

Date_Key
Course_Key
Professor_Key
Room_Key
Student_Key

Student attendance Fact Table

Time_Key
Product_Key
Store_Key
Promo_Key

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II. Dimensional Modeling and Star/Snowflake Schema

7. Why Need Multiple Fact Tables?

Multiple fact tables are needed in several scenarios like:

- value chain
- value circle
- heterogeneous product schemas
- Transaction and snapshot schemas
- Aggregates

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II. Dimensional Modeling and Star/Snowflake Schema

7. Why Need Multiple Fact Tables?

- Case of Value chain

Multiple fact tables are needed to support a business with many processes. Each process creates one or more fact tables. When these processes are arranged in order, they are called Value Chain.

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II. Dimensional Modeling and Star/Snowflake Schema

7. Why Need Multiple Fact Tables?

- For a Value chain
  - In Manufacturing business, a product travels through various steps, starting off as raw materials and ending as finished goods in the warehouse inventory. Steps include addition of ingredients, assembly of materials, process control, packaging etc.
  - From finished goods inventory, a product moves into shipment to distributor, distributor inventory, distributor shipment, retail inventory, …
  - At each step, value is added to the product. This workflow forms value chain or **supply chain**.

Other Examples are Insurance, Order Processing, and others.

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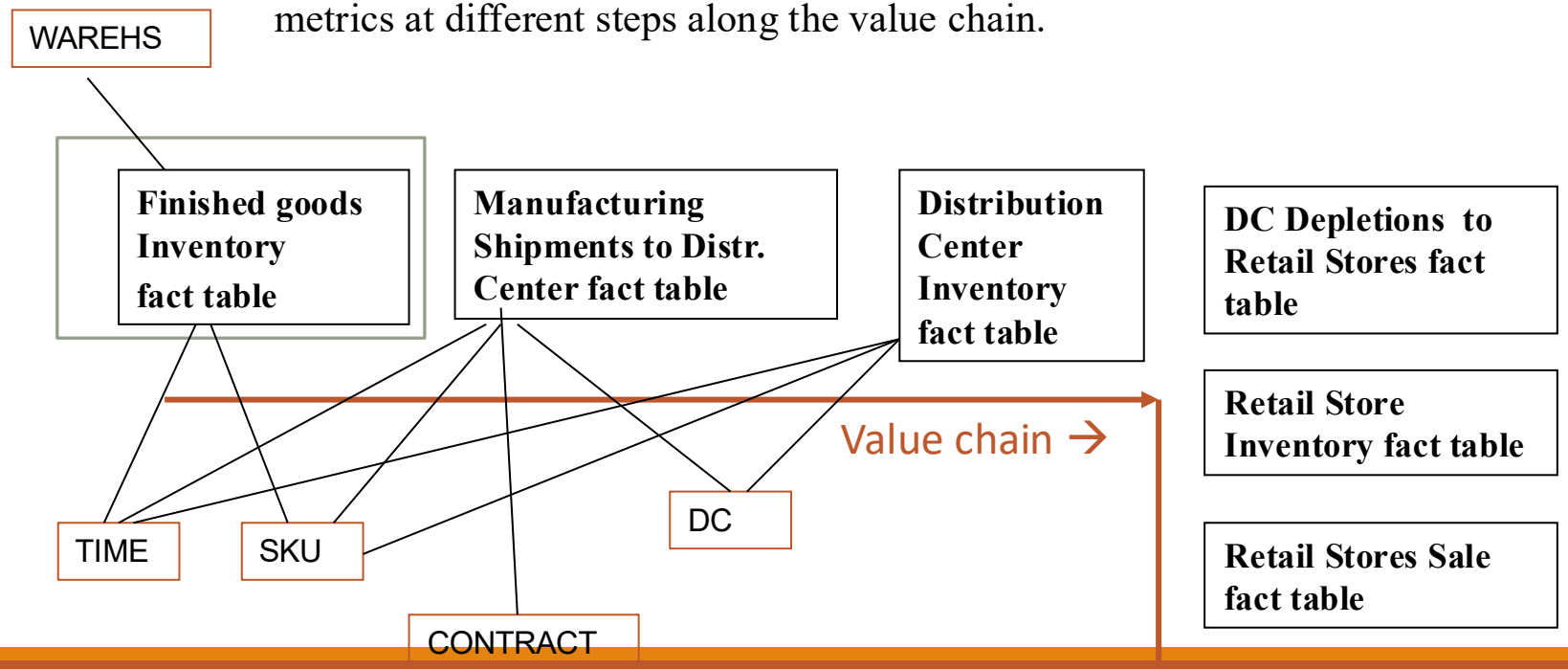
II. Dimensional Modeling and Star/Snowflake Schema

7. Why Need Multiple Fact Tables?

- A Typical Value chain
  - If you are in one of these businesses, you need to track important metrics at different steps along the value chain.



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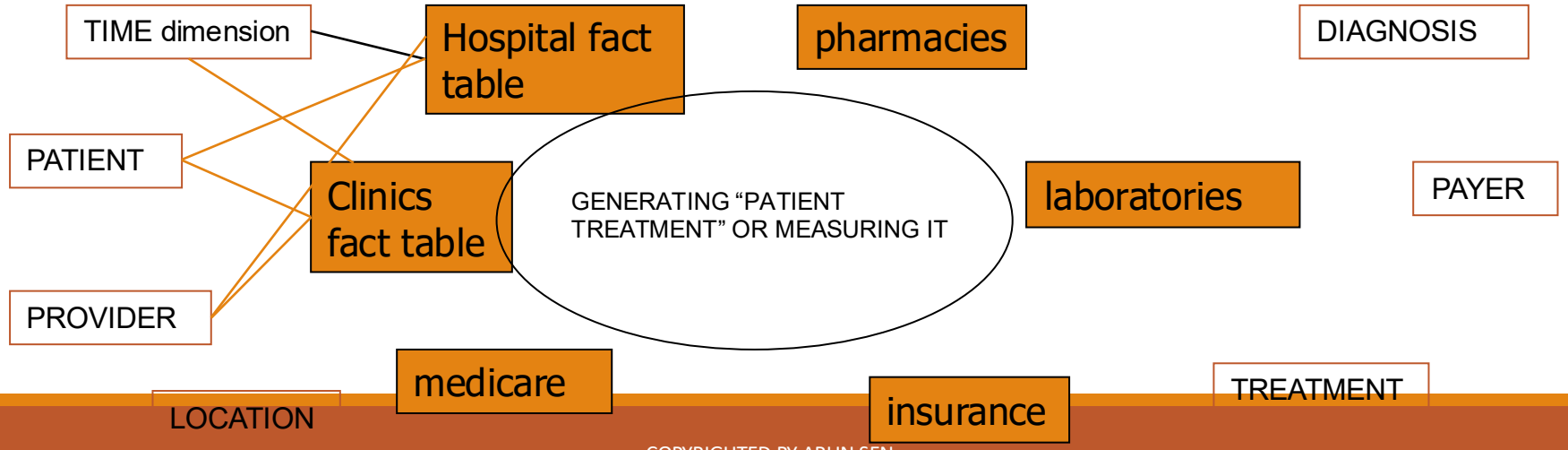
II. Dimensional Modeling and Star/Snowflake Schema

7. Why Need Multiple Fact Tables?

- For a Value circle
  - The businesses can be organized in a value circle so that they are performing or measuring the same kind of transaction. Examples are hospitals, clinics, physician offices, etc. for patient treatments.



II. Dimensional Modeling and Star/Snowflake Schema

7. Why Need Multiple Fact Tables?

- For a Heterogeneous product schema
  - There are many business that apparently "produce" things that are quite dissimilar. For example, a bank may have different lines of businesses, some of which are similar and others are dissimilar. The "savings a/c" and "checking a/c" are similar businesses while they are different from "credit card ops." These are called "**heterogeneous products.**" *How do you track these dissimilar services or products?*
    - *We use CORE FACT TABLE and CUSTOM FACT TABLE.*

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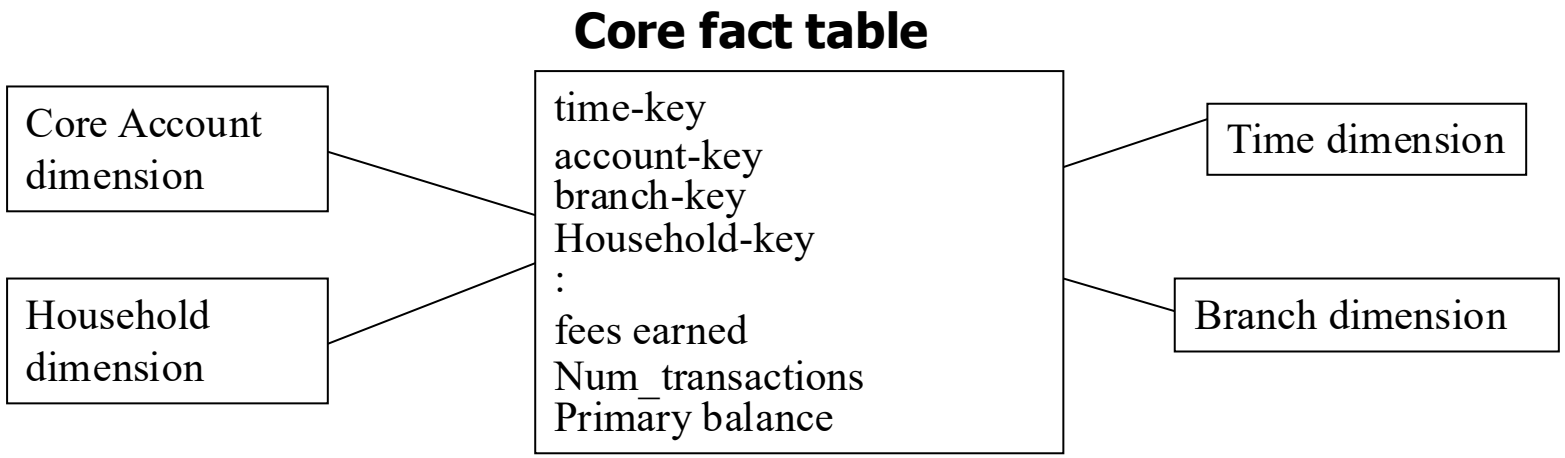
II. Dimensional Modeling and Star/Snowflake Schema

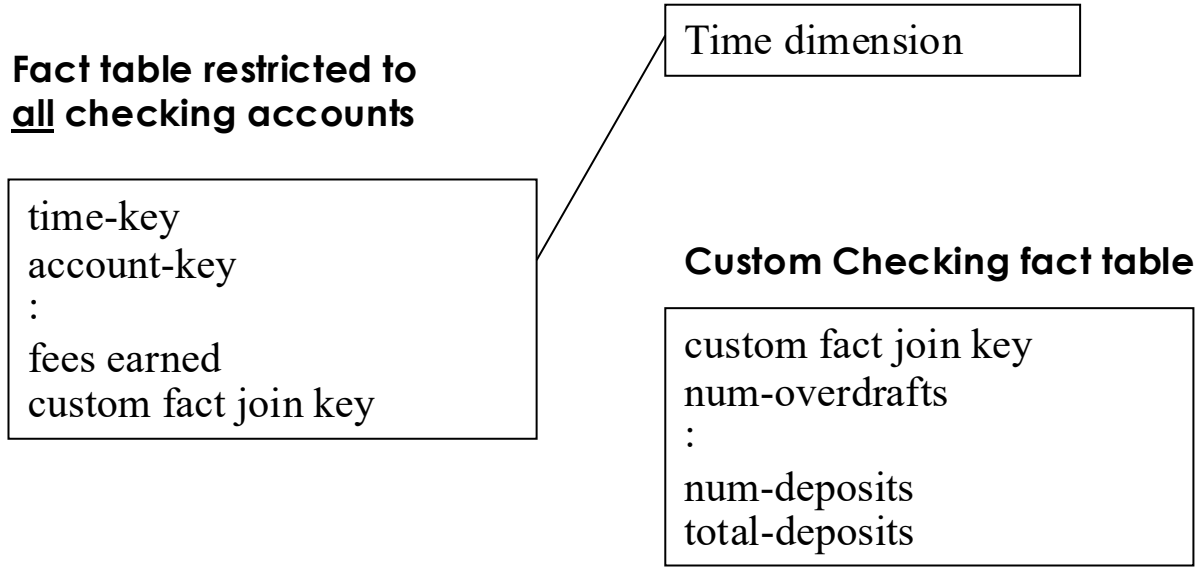
7. Why Need Multiple Fact Tables?



Bank Example: Global View of all accounts

II. Dimensional Modeling and Star/Snowflake Schema

7. Why Need Multiple Fact Tables?

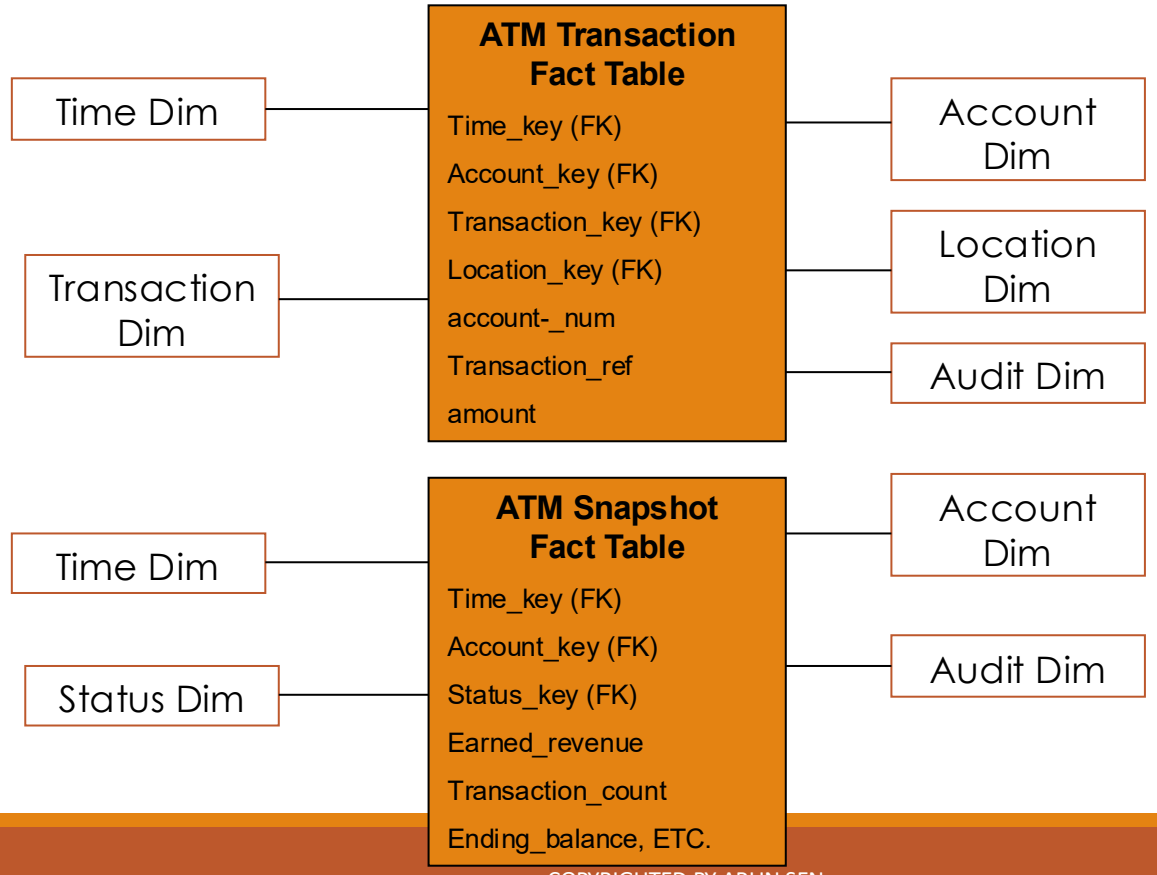


Bank Example: Line of Business View

II. Dimensional Modeling and Star/Snowflake Schema

7. Why Need Multiple Fact Tables?

- Transaction and Snapshot Schema Case



Models raw ATM transactions

Models periodic snapshots obtained from ATM transactions

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II. Dimensional Modeling and Star/Snowflake Schema

7. Why Need Multiple Fact Tables?

- Notion of Aggregates
  - Aggregates are stored summaries built primarily to improve query performance.
  - The creation of aggregates introduces a family of fact tables that are usually derived from the most granular fact table in each data mart.
  - Each member of the family represents a particular degree of summarization alone one or more of the fact table's dimensions.
  - A mature and full set of aggregates could involve a dozen separate aggregate fact tables, all echoing the original structure of the base fact table.

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II. Dimensional Modeling and Star/Snowflake Schema

8. What is a Dimension Table?

- Contains **descriptive** and **textual information**

- Attributes of Dimension Table are the source of most of constraints in the SQL Queries

- Examples of dimension are:
  a. product dimension
  b. store dimension
  c. customer dimension
  d. time dimension
  e. customer address - (commercial customers usually have hierarchies -- location to region)

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II. Dimensional Modeling and Star/Snowflake Schema

8. What is a Dimension Table?

Time Dimension

Time-key (PK)
Day-of-week
week-number
Month

Product Dim

Product-key (PK)
SKU
description
brand
category

Store Dimension

Store-key (PK)
store-ID
store-name
City
State
Region

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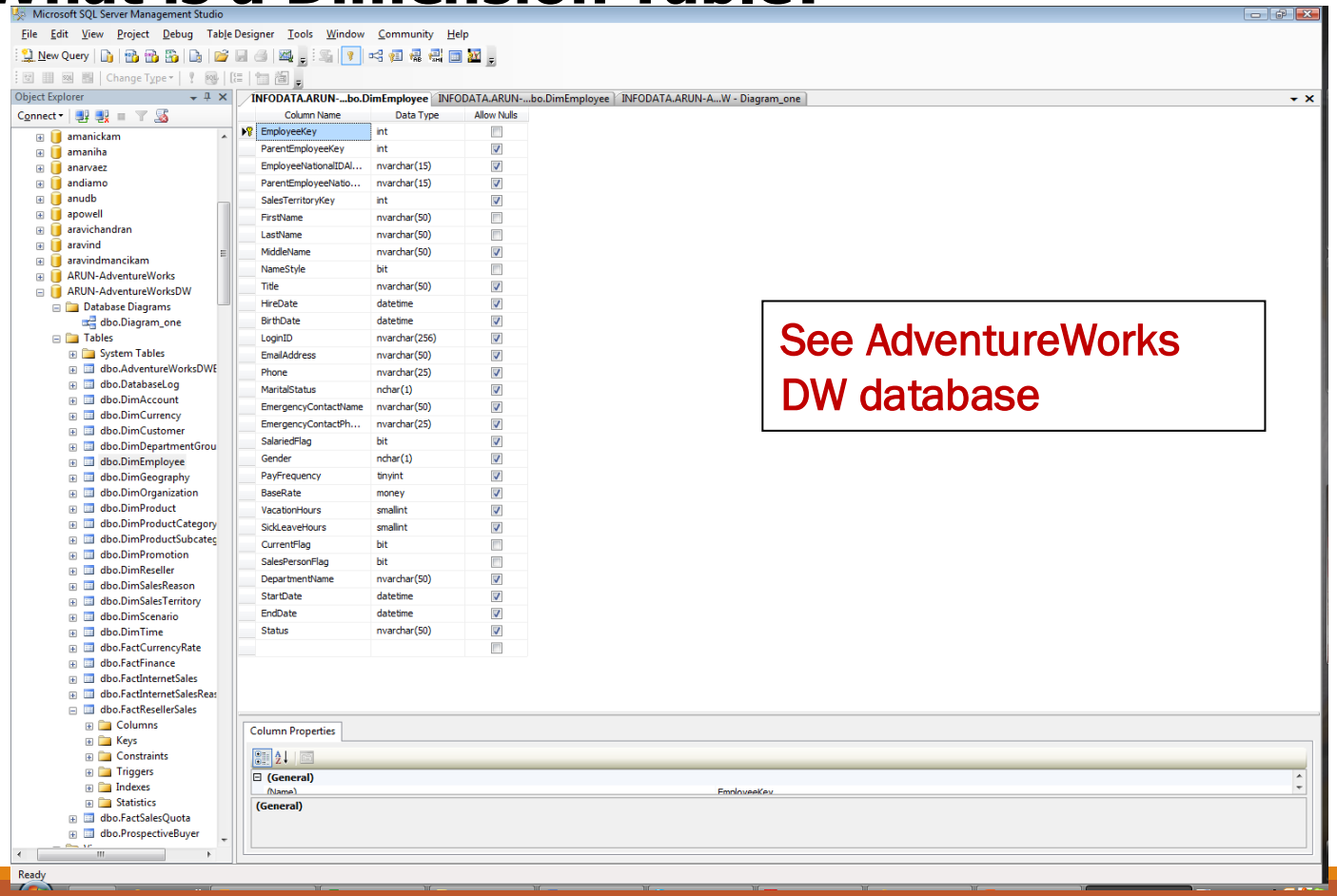
II. Dimensional Modeling and Star/Snowflake Schema

8. What is a Dimension Table?



See AdventureWorks DW database

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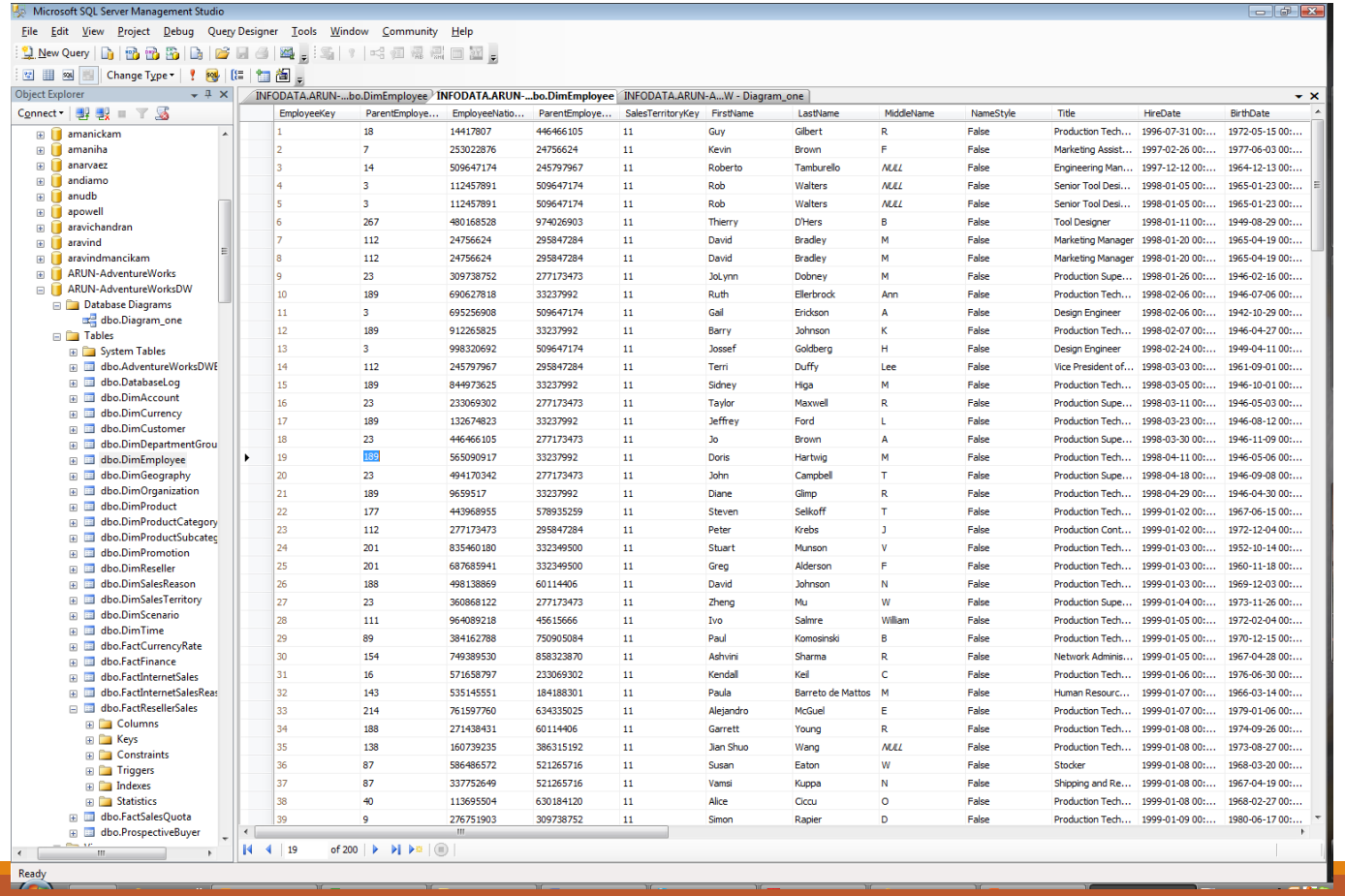
II. Dimensional Modeling and Star/Snowflake Schema

8. What is a Dimension Table?



EmployeeKey	ParentEmploye...	EmployeeNatio...	ParentEmploye...	SalesTerritoryKey	FirstName	LastName	MiddleName	NameStyle	Title	HireDate	BirthDate
1	18	14417807	446466105	11	Guy	Gilbert	R	False	Production Tech...	1996-07-31 00:...	1972-05-15 00:...
2	7	253022876	24756624	11	Kevin	Brown	F	False	Marketing Assist...	1997-02-26 00:...	1977-06-03 00:...
3	14	509647174	245797967	11	Roberto	Tamburello	NULL	False	Engineering Man...	1997-12-12 00:...	1964-12-13 00:...
4	3	112457891	509647174	11	Rob	Walters	NULL	False	Senior Tool Desi...	1998-01-05 00:...	1965-01-23 00:...
5	3	112457891	509647174	11	Rob	Walters	NULL	False	Senior Tool Desi...	1998-01-05 00:...	1965-01-23 00:...
6	267	480168528	974026903	11	Thierry	D'Hers	B	False	Tool Designer	1998-01-11 00:...	1949-08-29 00:...
7	112	24756624	295847284	11	David	Bradley	M	False	Marketing Manager	1998-01-20 00:...	1965-04-19 00:...
8	112	24756624	295847284	11	David	Bradley	M	False	Marketing Manager	1998-01-20 00:...	1965-04-19 00:...
9	23	309738752	277173473	11	JoLynn	Dobney	M	False	Production Supe...	1998-01-26 00:...	1946-02-16 00:...
10	189	690627818	33237992	11	Ruth	Ellerbrock	Ann	False	Production Tech...	1998-02-06 00:...	1946-07-06 00:...
11	3	695256908	509647174	11	Gail	Erickson	A	False	Design Engineer	1998-02-06 00:...	1942-10-29 00:...
12	189	912265825	33237992	11	Barry	Johnson	K	False	Production Tech...	1998-02-07 00:...	1946-04-27 00:...
13	3	998320692	509647174	11	Jossef	Goldberg	H	False	Design Engineer	1998-02-24 00:...	1949-04-11 00:...
14	112	245797967	295847284	11	Terri	Duffy	Lee	False	Vice President of...	1998-03-03 00:...	1961-09-01 00:...
15	189	844973625	33237992	11	Sidney	Higa	M	False	Production Tech...	1998-03-05 00:...	1946-10-01 00:...
16	23	233069302	277173473	11	Taylor	Maxwell	R	False	Production Supe...	1998-03-11 00:...	1946-05-03 00:...
17	189	132674823	33237992	11	Jeffrey	Ford	L	False	Production Tech...	1998-03-23 00:...	1946-08-12 00:...
18	23	446466105	277173473	11	Jo	Brown	A	False	Production Supe...	1998-03-30 00:...	1946-11-09 00:...
19	189	565090917	33237992	11	Doris	Hartwig	M	False	Production Tech...	1998-04-11 00:...	1946-05-06 00:...
20	23	494170342	277173473	11	John	Campbell	T	False	Production Supe...	1998-04-18 00:...	1946-09-08 00:...
21	189	9659517	33237992	11	Diane	Glimp	R	False	Production Tech...	1998-04-29 00:...	1946-04-30 00:...
22	177	443968955	578935259	11	Steven	Selikoff	T	False	Production Tech...	1999-01-02 00:...	1967-06-15 00:...
23	112	277173473	295847284	11	Peter	Krebs	J	False	Production Cont...	1999-01-02 00:...	1972-12-04 00:...
24	201	835460180	332349500	11	Stuart	Munson	V	False	Production Tech...	1999-01-03 00:...	1952-10-14 00:...
25	201	687685941	332349500	11	Greg	Alderson	F	False	Production Tech...	1999-01-03 00:...	1960-11-18 00:...
26	188	498138869	60114406	11	David	Johnson	N	False	Production Tech...	1999-01-03 00:...	1969-12-03 00:...
27	23	360868122	277173473	11	Zheng	Mu	W	False	Production Supe...	1999-01-04 00:...	1973-11-26 00:...
28	111	964089218	45615666	11	Ivo	Salmre	William	False	Production Tech...	1999-01-05 00:...	1972-02-04 00:...
29	89	384162788	750905084	11	Paul	Komosinski	B	False	Production Tech...	1999-01-05 00:...	1970-12-15 00:...
30	154	749389530	858323870	11	Ashvini	Sharma	R	False	Network Adminis...	1999-01-05 00:...	1967-04-28 00:...
31	16	571658797	233069302	11	Kendall	Keil	C	False	Production Tech...	1999-01-06 00:...	1976-06-30 00:...
32	143	535145551	184188301	11	Paula	Barreto de Mattos	M	False	Human Resourc...	1999-01-07 00:...	1966-03-14 00:...
33	214	761597760	634335025	11	Alejandro	McGuel	E	False	Production Tech...	1999-01-07 00:...	1979-01-06 00:...
34	188	271438431	60114406	11	Garrett	Young	R	False	Production Tech...	1999-01-08 00:...	1974-09-26 00:...
35	138	160739235	386315192	11	Jian Shuo	Wang	NULL	False	Production Tech...	1999-01-08 00:...	1973-08-27 00:...
36	87	586486572	521265716	11	Susan	Eaton	W	False	Stocker	1999-01-08 00:...	1968-03-20 00:...
37	87	337752649	521265716	11	Vamsi	Kuppa	N	False	Shipping and Re...	1999-01-08 00:...	1967-04-19 00:...
38	40	113695504	630184120	11	Alice	Ciccu	O	False	Production Tech...	1999-01-08 00:...	1968-02-27 00:...
39	9	276751903	309738752	11	Simon	Rapier	D	False	Production Tech...	1999-01-09 00:...	1980-06-17 00:...

See Adventure Works DW database

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II. Dimensional Modeling and Star/Snowflake Schema

8. What is a Dimension Table?

- Dimensions can be **lumped** to create more meaningful dimension such as

  <product, store> -- in case products are branded to sell in certain stores and also used for browsing

- Drilling Up and Down
  - Each dimension can have hierarchies (which could be more than one; also will talk about implementing this in SSAS) and browsers can be used to see the number of distinct values and to enumerate distinct relationships
  - "Drill down" means adding more details
  - "Drill up" means summarization or aggregation

.

II. Dimensional Modeling and Star/Snowflake Schema

9. Types of Dimension Table

- **Slowly Changing Dimension**
  - even if product key or production-customer key do not change the description of the product or customer does. To handle this problem, three options are available

    - **Type 1 Response:** Overwrite the dimension record with the new values. This is used when old description is not useful any more.

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II. Dimensional Modeling and Star/Snowflake Schema

9. Types of Dimension Table

- **Slowly Changing Dimension**
  - **Type 2 Response:** Create effective date attributes in the dimension table. This is used when a true physical change has taken place and it is appropriate to perfectly partition history by different descriptions. That is "Adding one row per change."

  - **Type 3 Response:** We do not have a change that perfectly partitions history. We create an "old" field in the dimension record to store the immediate previous attribute value. This is used when the sales district has occurred and the change is "soft."

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9. Types of Dimension Table

- **Rapidly Changing Small Dimension**
  - Rate structures continuously change in telecom industry
  - Use Type 2 of the above

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II. Dimensional Modeling and Star/Snowflake Schema

9. Types of Dimension Table

- **Large Dimensions**
  - With large granularity in data, we may end up with dimensions that are very large.
  - Examples: product dimension can contain "million" records; insurance companies can have "human" dimension that is very large along with "house" and "automobile" dimensions.
  - Although current RDBMS can handle such large tables, some conservative design approaches must be chosen.

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II. Dimensional Modeling and Star/Snowflake Schema

9. Types of Dimension Table

- **Rapidly Changing Monster Dimension**
  - Large Dimension has a tendency to change more likely than medium-sized one. For example, customer dimension can change more rapidly than the product dimension. This is because retailers may want to update customer data, etc. OR huge customer dimension with continuously revised demographics

- Solution is:
  - Break the dimension table (customer) into two tables (like customer and demographics)

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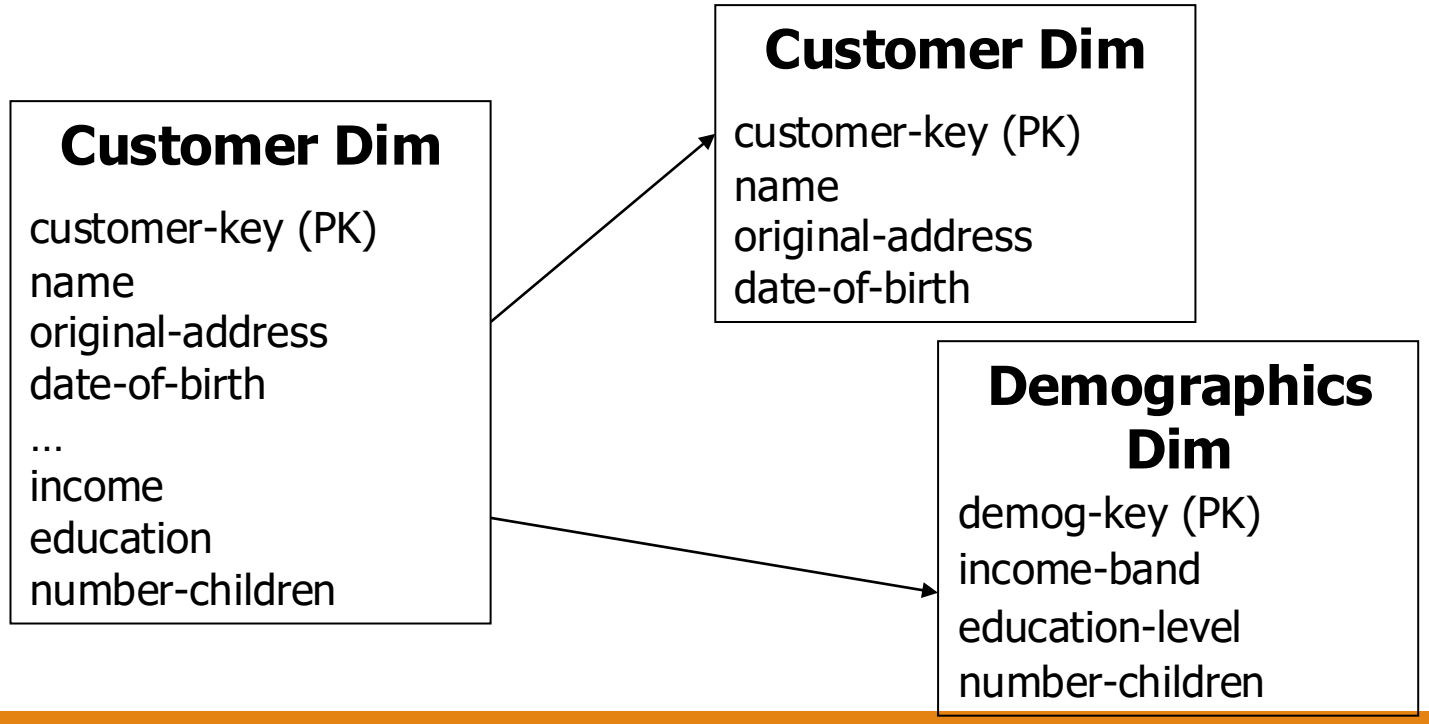
II. Dimensional Modeling and Star/Snowflake Schema

9. Types of Dimension Table

- Rapidly Changing Monster Dimension



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II. Dimensional Modeling and Star/Snowflake Schema

9. Types of Dimension Table

- However, there are some compromises:
  - attributes such as income, etc. must be converted to income_band
  - build demographics dimension with all possible discrete attribute combinations

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II. Dimensional Modeling and Star/Snowflake Schema

9. Types of Dimension Table

- **Junk dimension**
  - Sometimes you are left with flags and text attributes. These flags and text attributes do not organize in any coherent way. Often the meaning of these is quite obscure.
  - It is a bad idea
  - Choices:
    - Discard them – Not a good option as removing them may create problems for the business.
    - Leaving them in the fact table and they will swell the definition of fact table
    - Making each separate flag and attribute into its own dimension will increase the number of dimension!!

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II. Dimensional Modeling and Star/Snowflake Schema

9. Types of Dimension Table

- **Junk dimension**
  - Best will be ---
    - Create a dimension called Junk Dimension that is a convenient grouping of random flags and attributes to get them out of a fact table and into a useful dimensional framework.

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II. Dimensional Modeling and Star/Snowflake Schema

10. Different Keys used in Dimensional Modeling

- **Foreign Keys** (mostly reside in fact tables; can appear in dimension table also in snowflake schema)
- **Primary Keys** (same notion as we know; appears mostly in dimension tables)
- **Surrogate Keys** (These are DW Keys and are meaningless. They are used to connect dimension tables with the fact table. Usually, it is a 4-byte integer. Also applicable to Dates.)

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II. Dimensional Modeling and Star/Snowflake Schema

10. Different Keys used in Dimensional Modeling

- Avoid Smart Keys (Avoid the temptation of making overloaded keys.)
- Do not use production keys (These are the PKs in the production process.)

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II. Dimensional Modeling and Star/Snowflake Schema

11. Kimball Methodology to Build Star/Snowflake Schema using

Step-1: Requirements Analysis as discussed earlier

- **get business questions**
- **look for business objectives and issues**
- **collect analysis requirements**

Step-2: Build the Matrix

- The best way to build the schema is to create a matrix with **data marts** in one axis (x-axis) and **dimensions** in the other axis (y-axis).
- We define a data mart as a pragmatic collection of related facts.

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II. Dimensional Modeling and Star/Snowflake Schema

11. Steps to Build Star/Snowflake Schema (Kimball)

Step 2.1. List the Data Marts in the Matrix

- Choose single-source data marts first. Single-source data marts involve single kind of data source.
- This is easier to handle. Also, you will be able to deliver something worthwhile to the end users quickly.
- Single-source data marts are *customer billing, customer inquiries, promotion tracking* [Each uses several dimensions <u>but</u> come from same source data]
- Create multiple-source data marts from the single-source data marts, if possible.
- Multiple-source data marts are *customer relationship management using customer billing, customer inquiries* and *promotion tracking single-source data marts*.

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II. Dimensional Modeling and Star/Snowflake Schema

11. Steps to Build Star/Snowflake Schema (Kimball)

Step 2.2. List the Dimensions in the Matrix

Step 2.3. Mark the intersection in the Matrix

- Matrix is usually very dense
- Each data mart has on the average 10 dimensions.

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II. Dimensional Modeling and Star/Snowflake Schema

11. Steps to Build Star/Snowflake Schema (Kimball)

Dimensions (will be used as dimension tables later) → Data Marts ↓	Time	Customer	Service	Rate Category	Local Service provider	Calling Party	Called Party	Long-distance Provider	Internal Organization	Employee	Location		
customer billing	x	x	x	x	x			x			x		
service orders	x	x	x		x			x	x	x	x		
toruble reports	x	x	x		x	x		x	x	x	x		
customer inquires	x	x	x	x	x	x		x	x	x	x		
bill call detail	x	x	x	x	x	x	x	x	x		x		
network call detail	x	x	x	x	x	x	x	x	x		x		
purchase orders	x								x	x	x		
supplier deliveries	x								x	x	x		

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II. Dimensional Modeling and Star/Snowflake Schema

11. Steps to Build Star/Snowflake Schema (Kimball)

- One data mart can overlap with another data mart.
- Go for 10-30 data marts for a large organization.
- In case you go with 30-40 data marts, you have probably divided the problem down to too fine a grain.

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II. Dimensional Modeling and Star/Snowflake Schema

11. Steps to Build Star/Snowflake Schema (Kimball)

Step 3: Design Fact Tables

Step 3A. Choose the Data Mart

- Look down the row headings of our matrix and choose one of the data marts. Go for single-source data mart first.

Step 3B. Declare the Grain

- Declaring the grain means declaring what is an individual fact record.

Step 3C. Choose the Facts Based Upon the Grain

- Remember the things that we discussed about facts, their additive, semi-additive and non-additive properties

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II. Dimensional Modeling and Star/Snowflake Schema

11. Steps to Build Star/Snowflake Schema (Kimball)

Step 3D. Include Derived Facts

- Some facts in the fact table are derived such as YTD sales, Gross Profit, etc. Keep track of them.
- There are two kinds of **derived facts**. First, derived facts can be additive. Easy stuff. Two, derived facts can be a nonadditive calculation such as Ratio.
- **Remember: Include your derived facts at the time you are creating other facts. Do not wait till you reach end user applications.**

Sales Fact Table

Time-key (FK)
Product-key (FK)
Store-key (FK)
dollars-sold
units-sold
dollars-cost
YTD dollars-cost

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II. Dimensional Modeling and Star/Snowflake Schema

11. Steps to Build Star/Snowflake Schema (Kimball)

Step 3E. Follow the following Steps for Base Facts and Derived Facts

- Identify all possible base and derived facts. Collect all reports submitted during the requirements analysis phase.
- Create a spreadsheet of facts and derived facts along with the name of person who provided it, etc.
- Remove duplicate facts
- Discover and document the underlying calculations
- Cross reference base facts
- Get approval for the derived facts

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II. Dimensional Modeling and Star/Snowflake Schema

11. Steps to Build Star/Snowflake Schema (Kimball)

Step 3F. Create Fact Table Diagram

- It is very worthwhile to create a logical diagram for each fact table constructed along with its dimensions. This will not only show the context of the fact table but also the context of the fact table with respect to the data mart.
- The diagram can be used to: (A) communicate with the end users; (B) help select data access tool selection; and (C) end user training.

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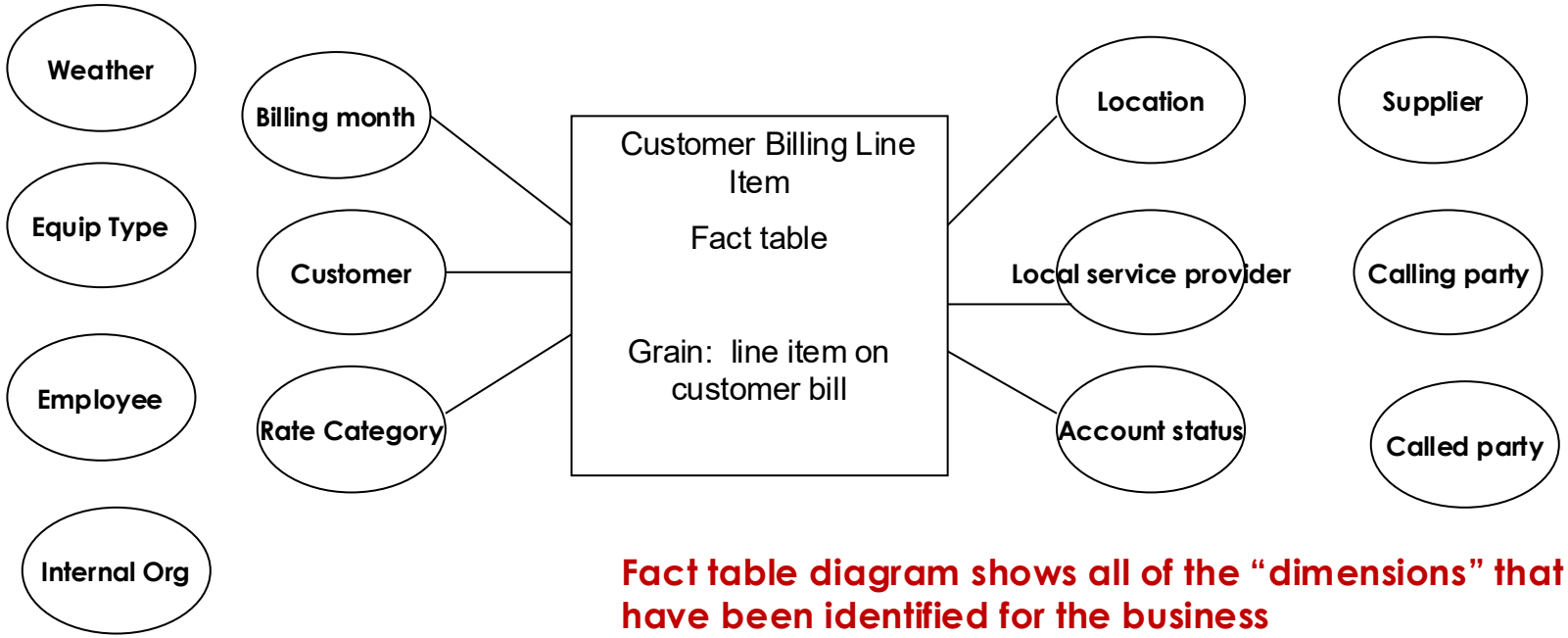
11. Steps to Build Star/Snowflake Schema (Kimball)

Step 3F. Create Fact Table Diagram



Fact table diagram shows all of the "dimensions" that have been identified for the business

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II. Dimensional Modeling and Star/Snowflake Schema

11. Steps to Build Star/Snowflake Schema (Kimball)

Step 4: Design the Dimension Tables

- We draw a **dimension table diagram** for each dimension
- The diagram shows explicit grain of each dimension.
- It also shows the approximate cardinality of each dimension attribute.
- It allows the user to quickly see the multiple hierarchies and relationships between the attributes. This shows the default drill path.
- Full descriptive data must be provided to support the above diagram

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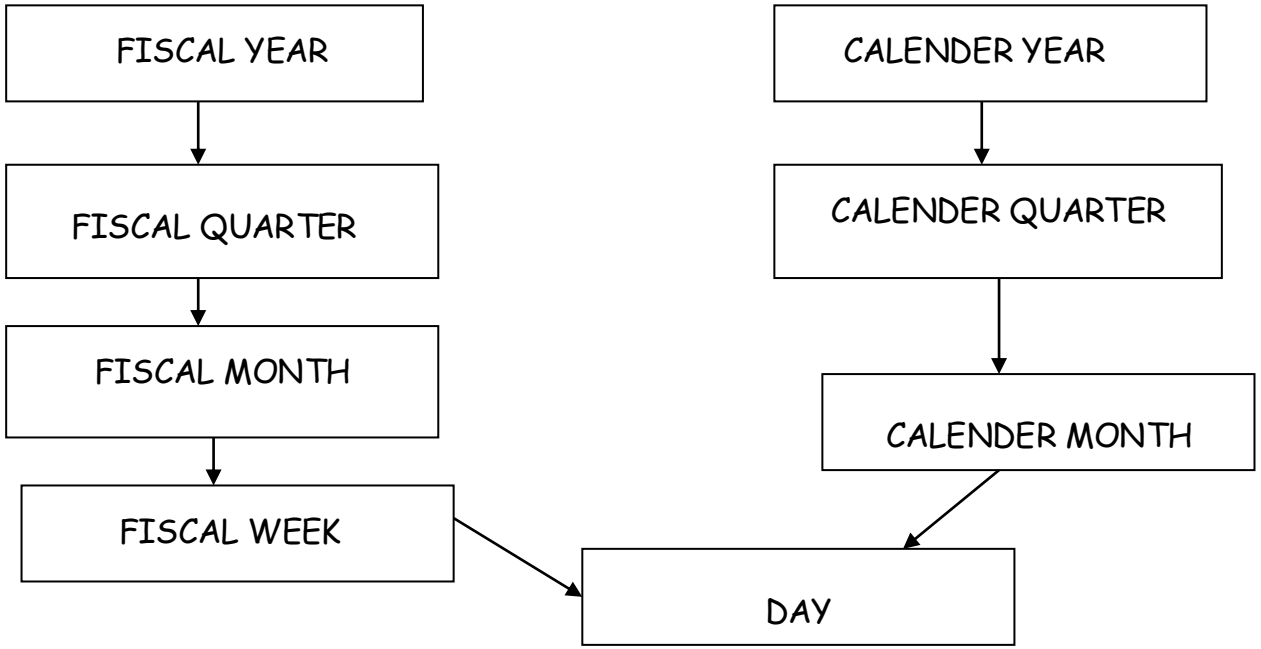
11. Steps to Build Star/Snowflake Schema (Kimball)

Step 4: Design the Dimension Tables



II. Dimensional Modeling and Star/Snowflake Schema

11. Steps to Build Star/Snowflake Schema (Kimball)

Step 5: Feedback for the Design (Fact And Dimension Tables)

Step 5 A. Get Inputs from the IS Team

- Focus: Need to see what are missed in the current dimensional model

Step 5 B. Get Inputs from the Core Business Users

- Focus: (a) See if the dimension model helps all the types of reports the core business users are envisioning to have; (b) Navigation through the large dimensions

Step 5 C. Present to Business Users

- Focus: (a) check the model against the business questions;

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II. Dimensional Modeling and Star/Snowflake Schema

11. Steps to Build Star/Snowflake Schema (Kimball)

Step 6: Data Sourcing

- There are two kinds of data sources: informal and formal.
  - Informal data sources are personal databases and new data input. In both cases, **make sure** to clean the data before you let them in the data warehouse.
  - Formal sources are usually supported by the database staff

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II. Dimensional Modeling and Star/Snowflake Schema

11. Steps to Build Star/Snowflake Schema (Kimball)

Step 6: Data Sourcing

- Create Data Source Definitions
  - source -- name of the source system
  - business owner -- the name of the primary contact responsible
  - IS owner -- the name of the person who is responsible for the source system
  - platform
  - location
  - description

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II. Dimensional Modeling and Star/Snowflake Schema

11. Steps to Build Star/Snowflake Schema (Kimball)

Step 6: Data Sourcing

- Remember that there will be external data providers (could be data vendors, or could be from other divisions of the organization)
- Facts need to come from one data source system
- **Data Extraction Planning**
  - Data Accessibility -- pick the easier access data source
  - Look for the step that feeds the data
  - Data Accuracy
- **Mapping Data from Source to Staging (type 1)**

Source data	Source data field	Mapping	Staging Table type	Staging Table name	attribute
Sales.csv	Customer-name	Copy	Relation	Sales_staging	C-name

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II. Dimensional Modeling and Star/Snowflake Schema

11. Steps to Build Star/Snowflake Schema (Kimball)

Step 6: Data Sourcing

- **Data Loading from Staging to Presentation Server**

  You need to load the data from staging to presentation server after doing all the transformations

- **Mapping Data from Staging to Presentation Server (type 2)**

Source data in staging	Staging table data field	Mapping	Data Mart Table type	Table name	attribute
Sales_staging table	C-name	Copy	Dimension Table	Dim.customer	CustName

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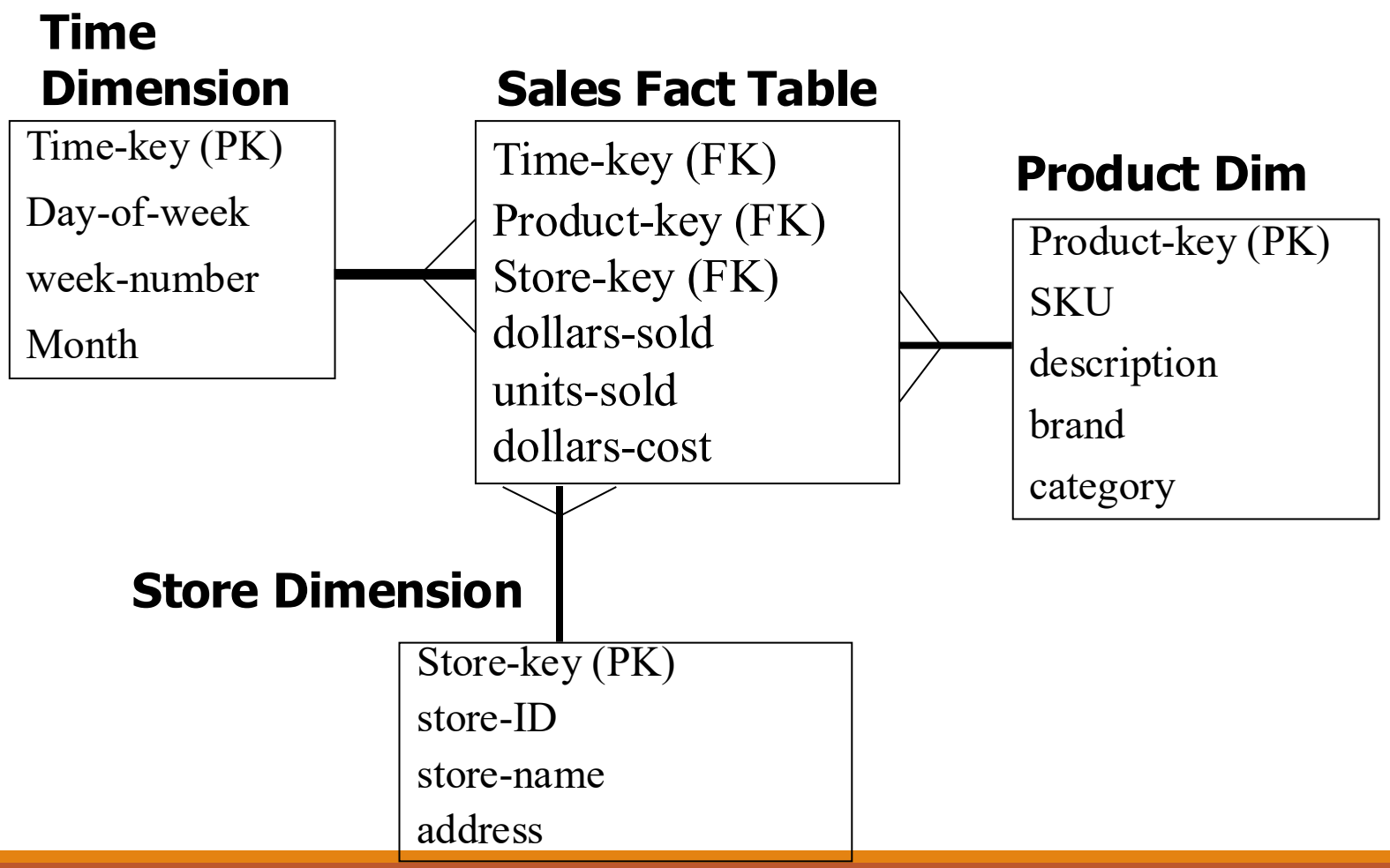
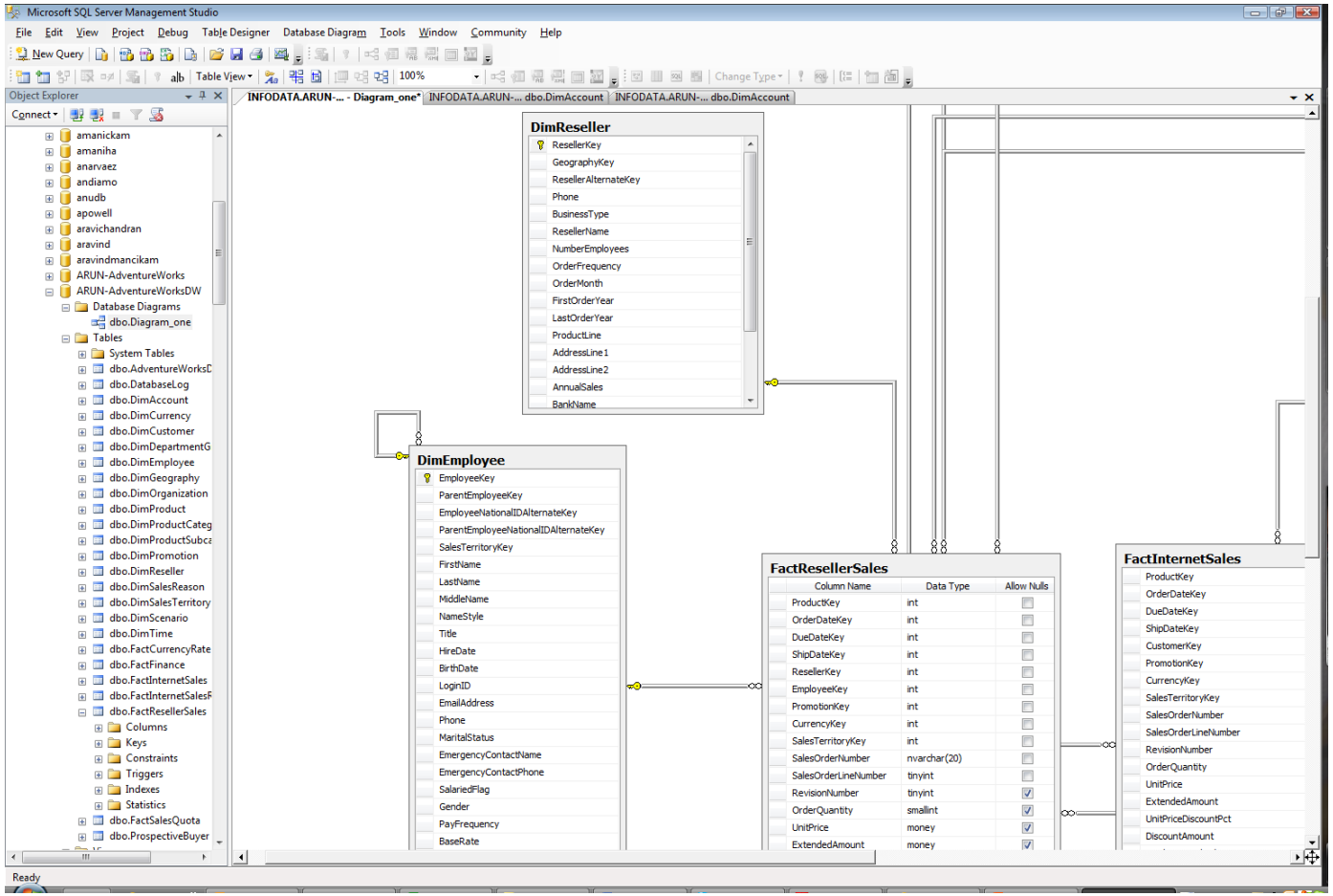
Star Schema


Time Dimension
Time-key (PK)
Day-of-week
week-number
Month
Sales Fact Table
Time-key (FK)
Product-key (FK)
Store-key (FK)
dollars-sold
units-sold
dollars-cost
Product Dim
Product-key (PK)
SKU
description
brand
category
Store Dimension
Store-key (PK)
store-ID
store-name
address

Star Schema



http://www.youtube.com/watch?v=A-G5UVohwQQ

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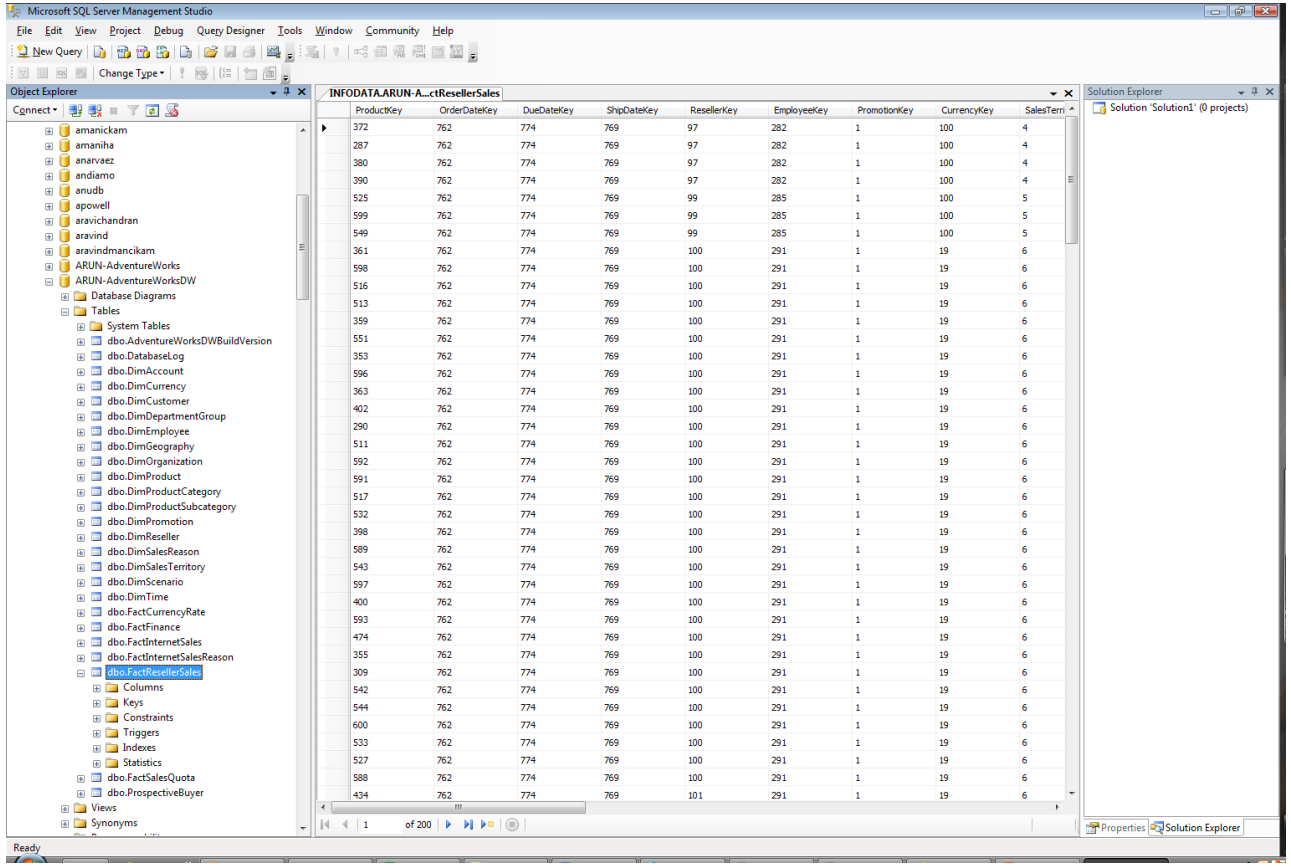
II. Dimensional Modeling and Star/Snowflake Schema

Fact Table



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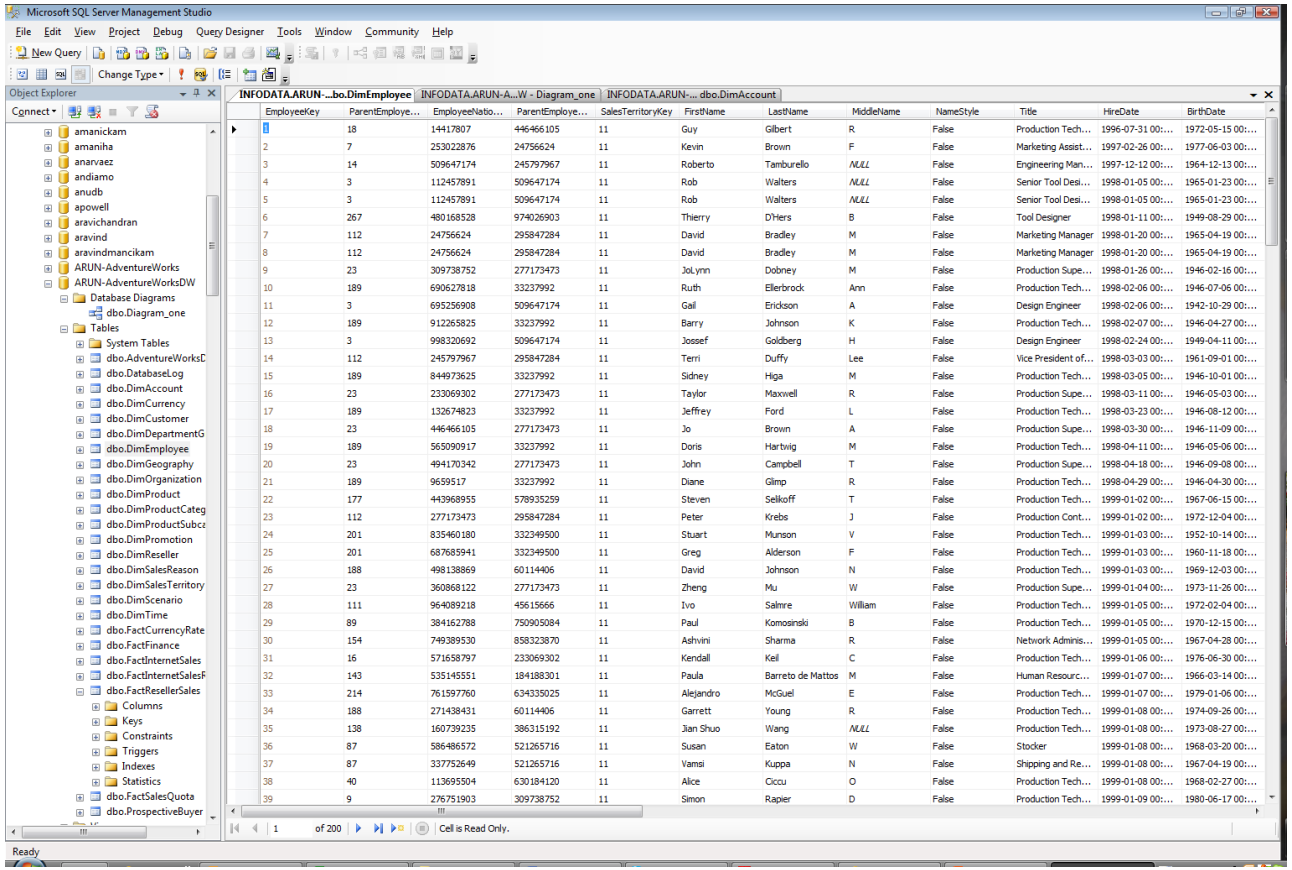
II. Dimensional Modeling and Star/Snowflake Schema

dimEmployee Table



EmployeeKey	ParentEmploye...	EmployeeNatio...	ParentEmploye...	SalesTerritoryKey	FirstName	LastName	MiddleName	NameStyle	Title	HireDate	BirthDate
1	18	14417807	446466105	11	Guy	Gilbert	R	False	Production Tech...	1996-07-31 00:...	1972-05-15 00:...
2	7	253022876	24756624	11	Kevin	Brown	F	False	Marketing Assist...	1997-02-26 00:...	1977-06-03 00:...
3	14	509647174	245797967	11	Roberto	Tamburello	NULL	False	Engineering Man...	1997-12-12 00:...	1964-12-13 00:...
4	3	112457891	509647174	11	Rob	Walters	NULL	False	Senior Tool Desi...	1998-01-05 00:...	1965-01-23 00:...
5	3	112457891	509647174	11	Rob	Walters	NULL	False	Senior Tool Desi...	1998-01-05 00:...	1965-01-23 00:...
6	267	480168528	974026903	11	Thierry	D'Hers	B	False	Tool Designer	1998-01-11 00:...	1949-08-29 00:...
7	112	24756624	295847284	11	David	Bradley	M	False	Marketing Manager	1998-01-20 00:...	1965-04-19 00:...
8	112	24756624	295847284	11	David	Bradley	M	False	Marketing Manager	1998-01-20 00:...	1965-04-19 00:...
9	23	309738752	277173473	11	JoLynn	Dobney	M	False	Production Supe...	1998-01-26 00:...	1946-02-16 00:...
10	189	690627818	33237992	11	Ruth	Ellerbrock	Ann	False	Production Tech...	1998-02-06 00:...	1946-07-06 00:...
11	3	695256908	509647174	11	Gail	Erickson	A	False	Design Engineer	1998-02-06 00:...	1942-10-29 00:...
12	189	912265825	33237992	11	Barry	Johnson	K	False	Production Tech...	1998-02-07 00:...	1946-04-27 00:...
13	3	998320692	509647174	11	Jossef	Goldberg	H	False	Design Engineer	1998-02-24 00:...	1949-04-11 00:...
14	112	245797967	295847284	11	Terri	Duffy	Lee	False	Vice President of...	1998-03-03 00:...	1961-09-01 00:...
15	189	844973625	33237992	11	Sidney	Higa	M	False	Production Tech...	1998-03-05 00:...	1946-10-01 00:...
16	23	233069302	277173473	11	Taylor	Maxwell	R	False	Production Supe...	1998-03-11 00:...	1946-05-03 00:...
17	189	132674823	33237992	11	Jeffrey	Ford	L	False	Production Tech...	1998-03-23 00:...	1946-08-12 00:...
18	23	446466105	277173473	11	Jo	Brown	A	False	Production Supe...	1998-03-30 00:...	1946-11-09 00:...
19	189	565090917	33237992	11	Doris	Hartwig	M	False	Production Tech...	1998-04-11 00:...	1946-05-06 00:...
20	23	494170342	277173473	11	John	Campbell	T	False	Production Supe...	1998-04-18 00:...	1946-09-08 00:...
21	189	9659517	33237992	11	Diane	Glimp	R	False	Production Tech...	1998-04-29 00:...	1946-04-30 00:...
22	177	443968955	578935259	11	Steven	Selkoff	T	False	Production Tech...	1999-01-02 00:...	1967-06-15 00:...
23	112	277173473	295847284	11	Peter	Krebs	J	False	Production Cont...	1999-01-02 00:...	1972-12-04 00:...
24	201	835460180	332349500	11	Stuart	Munson	V	False	Production Tech...	1999-01-03 00:...	1952-10-14 00:...
25	201	687685941	332349500	11	Greg	Alderson	F	False	Production Tech...	1999-01-03 00:...	1960-11-18 00:...
26	188	498138869	60114406	11	David	Johnson	N	False	Production Tech...	1999-01-03 00:...	1969-12-03 00:...
27	23	360868122	277173473	11	Zheng	Mu	W	False	Production Supe...	1999-01-04 00:...	1973-11-26 00:...
28	111	964089218	45615666	11	Ivo	Salmre	William	False	Production Tech...	1999-01-05 00:...	1972-02-04 00:...
29	89	384162788	750905084	11	Paul	Komosinski	B	False	Production Tech...	1999-01-05 00:...	1970-12-15 00:...
30	154	749389530	858323870	11	Ashvini	Sharma	R	False	Network Adminis...	1999-01-05 00:...	1967-04-28 00:...
31	16	571658797	233069302	11	Kendall	Keil	C	False	Production Tech...	1999-01-06 00:...	1976-06-30 00:...
32	143	535145551	184188301	11	Paula	Barreto de Mattos	M	False	Human Resourc...	1999-01-07 00:...	1966-03-14 00:...
33	214	761597760	634335025	11	Alejandro	McGuel	E	False	Production Tech...	1999-01-07 00:...	1979-01-06 00:...
34	188	271438431	60114406	11	Garrett	Young	R	False	Production Tech...	1999-01-08 00:...	1974-09-26 00:...
35	138	160739235	386315192	11	Jian Shuo	Wang	NULL	False	Production Tech...	1999-01-08 00:...	1973-08-27 00:...
36	87	586486572	521265716	11	Susan	Eaton	W	False	Stocker	1999-01-08 00:...	1968-03-20 00:...
37	87	337752649	521265716	11	Vamsi	Kuppa	N	False	Shipping and Re...	1999-01-08 00:...	1967-04-19 00:...
38	40	113695504	630184120	11	Alice	Ciccu	O	False	Production Tech...	1999-01-08 00:...	1968-02-27 00:...
39	9	276751903	309738752	11	Simon	Rapier	D	False	Production Tech...	1999-01-09 00:...	1980-06-17 00:...

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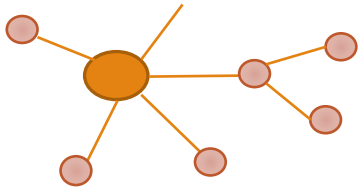
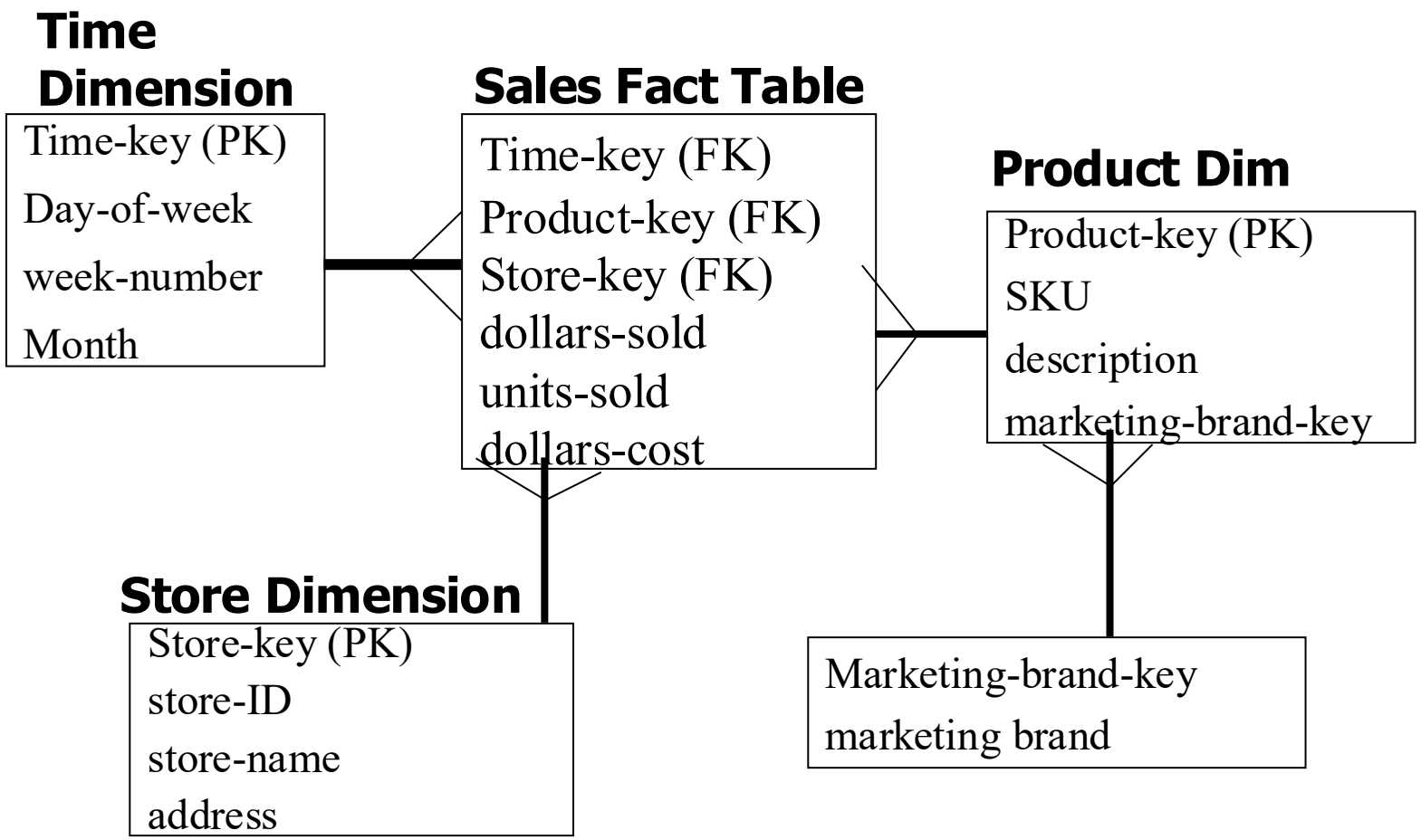
Snowflake Schema





Snowflake Schema

12. Permissible Snowflaking

- A dimension is said to be "snowflaked" when the low cardinality fields in the dimension have been removed to separate tables and linked back into the original table with artificial key. This helps if the user is asking only for the distinct values of the category….
- Snowflaking makes most forms of browsing among the dimension attributes slower.
- Not always a good idea because --
  - creates confusion with the users' understanding
  - create problems with "bit mapped indexes"
  - more administration cost

(Low-cardinality refers to columns with few unique values. Low-cardinality column values are typically status flags, Boolean values, or major classifications such as gender. An example of a data table column with low-cardinality would be a CUSTOMER table with a column named NEW_CUSTOMER. This column would contain only 2 distinct values: Y or N, denoting whether the customer was new or not. Since there are only 2 possible values held in this column, its cardinality type would be referred to as low-cardinality. – Wikipedia)

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Examples from a TELCO Case Study (Kimball)

Dimension Tables

Date Dimension

DATE_KEY
day
day_no_in_cal_month
day_no_in cal_year
day_no_in_fiscal_mo
day_no_in_fiscal_year
day_of_week
holiday_flag
type_of_day
cal_week
cal_week_no
cal_month
cal_month_no
cal_quarter
cal_quarter_no
cal_year
fiscal_week
fiscal_week_no
fiscal_month
fiscal_month_no
fiscal_quarter
fiscal_quarter_no
fiscal_year

Billing Date Dimension

BILLING_DATE_KEY
billing_month
billing_month_no
billing_quarter
billing_quarter_no
billing_year

End Date Dimension

END_DATE_KEY
end_day
end_day_no_in_cal_month
end_day_no_in cal_year
end_day_no_in_fiscal_mo
end_day_no_in_fiscal_year
end_day_of_week
end_holiday_flag
end_type_of_day
end_cal_week
end_cal_week_no
end_cal_month
end_cal_month_no
end_cal_quarter
end_cal_quarter_no
end_cal_year
end_fiscal_week
end_fiscal_week_no
end_fiscal_month
end_fiscal_month_no
end_fiscal_quarter
end_fiscal_quarter_no
end_fiscal_year

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Dimension Tables

Time Dimension

TIME_KEY
minute
hour
am_pm_flag
daypart

Customer Dimension

CUSTOMER_KEY
deliv_pt_id
deliv_pt_descr
deliv_pt_suffix
deliv_pt_prefix
deliv_pt_area_code
deliv_pt_type
cust_loc_id
cust_loc_descr
assigned_service_region
cust_loc_street_no
cust_loc_street_name
cust_loc_street_type
cust_loc_street_direction
cust_loc_po_box
cust_loc_suite
cust_loc_city
cust_loc_state_province
cust_loc_country
cust_loc_5_difgit_zip
cust_loc_9_digit_zip

(continued)

customer_id
customer name
cust_type
assigned_sales_region
date_of_1st_service
cust_street_no
cust_street_name
cust_street_type
cust_street_direction
cust_po_box
cust_suite
cust_city
cust_state_province
cust_country
cust_5_digit_zip
cust_9_digit_zip
cust_gender
cust_date_of_birth
cust_demogr_profile
cust_educ_level
household_id
household_profile
cust_industry
cust_industry_group
cust_parent company

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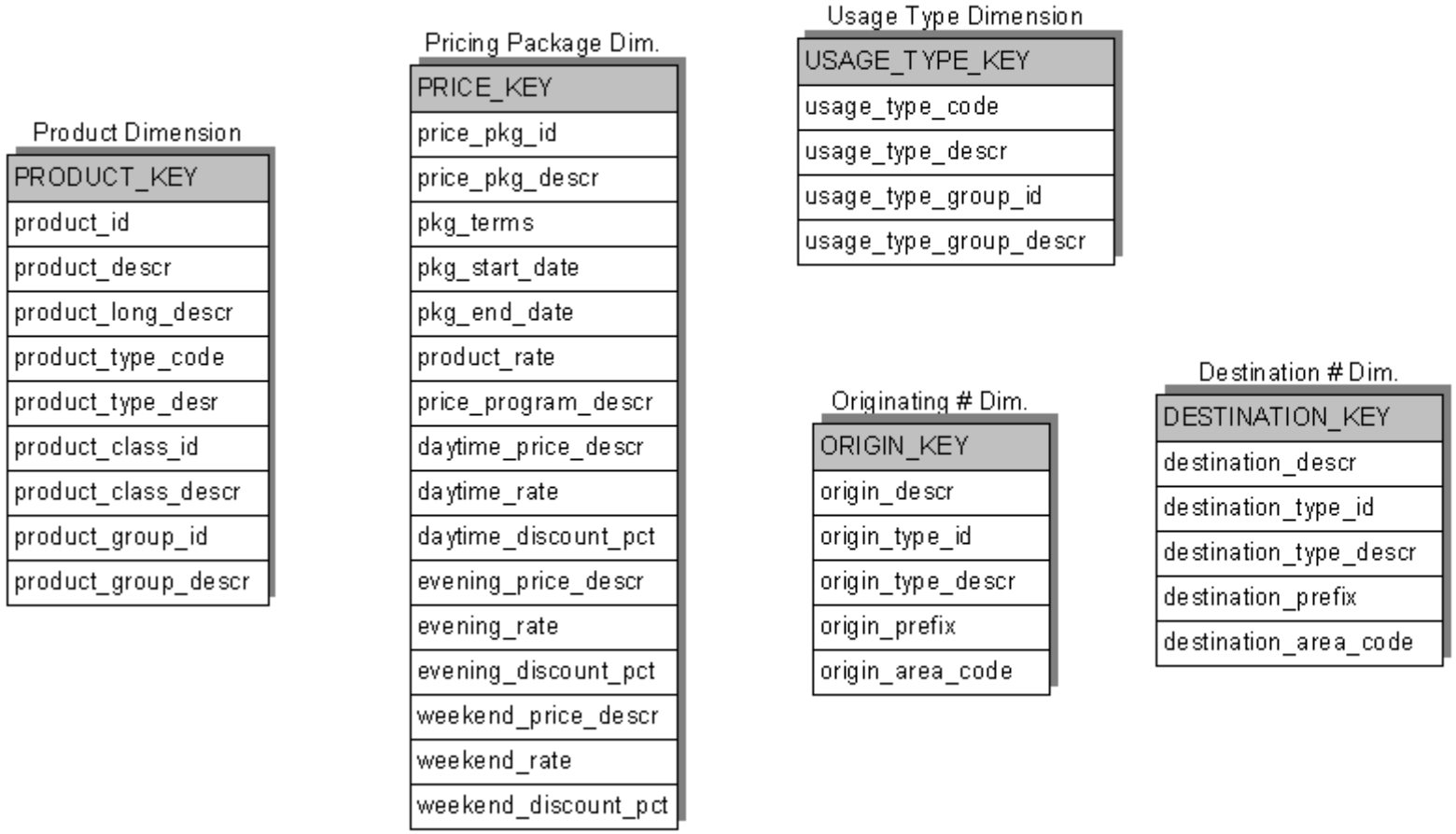
Dimension Tables



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Dimension Tables

Vendor Dimension

VENDOR_KEY
vendor_loc_id
vendor_loc_descr
vendor_loc_city
vendor_loc_state_province
vendor_loc_country
vendor_loc_latitude
vendor_loc_longitude
vendor_code
vendor_descr
vendor_long_descr
minor_ownership_flag
vendor_status
vendor_parent_company
vendor_street_no
vendor_street_name
vendor_street_type
vendor_street_direction
vendor_po_box
vendor_suite
vendor_city
vendor_state_province
vendor_country

Purchased Product Dim.

PURCH_PROD_KEY
purch_prod_code
purch_prod_descr
purch_subcat_code
purch_subcat_descr
purch_category_code
purch_category_descr
buyer_id
buyer_name
purch_prod_group_code
pruch_prod_group_descr

GL Account Dimension

GL_ACCOUNT_KEY
gl_account_id
gl_account_descr
account_type
account_group3_id
account_group3_descr
account_group2_id
account_group2_descr
account_group1_id
account_group1_descr

Organization Dimension

ORG_KEY
cost_center_code
cost_center_descr
bus_unit_code
bus_unit_descr
region_code
region_descr
division_code
division_descr
company_code
company_descr
cost_center_city
cost_center_state_province
cost_center_country

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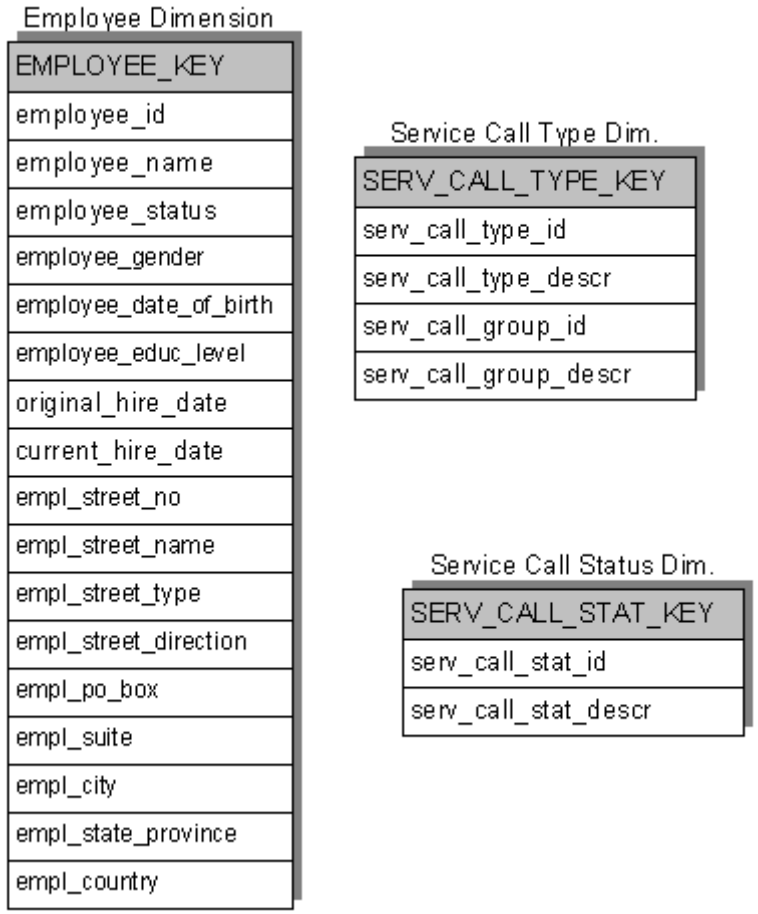
Dimension Tables



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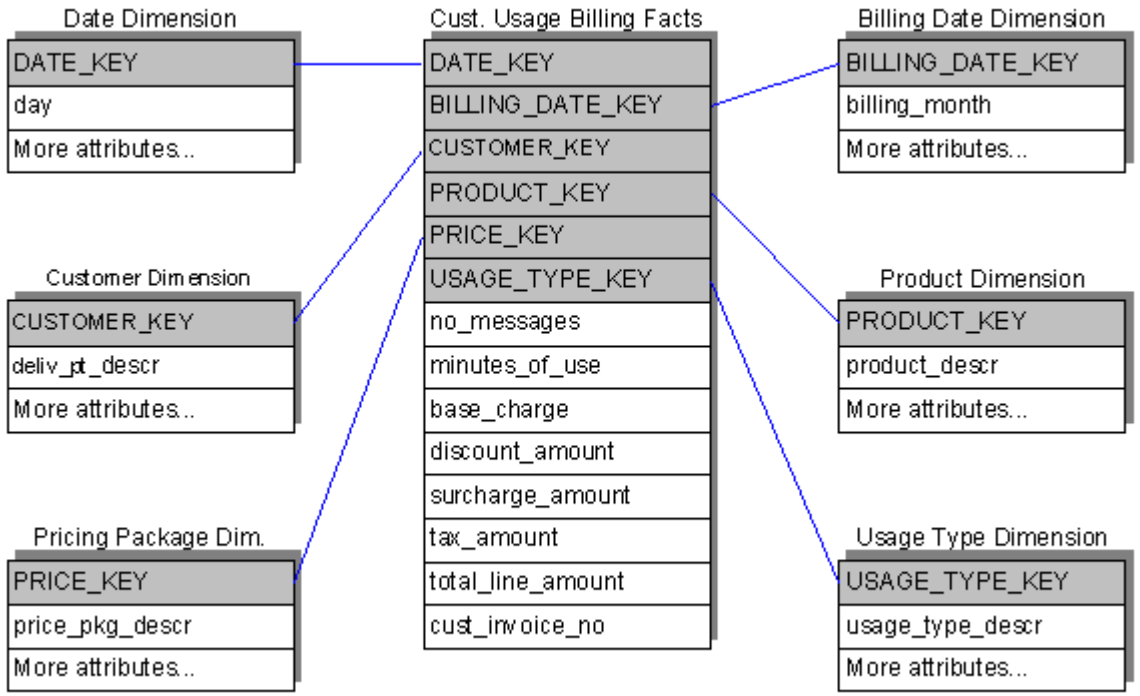
Star Schema *Customer Usage Billing*



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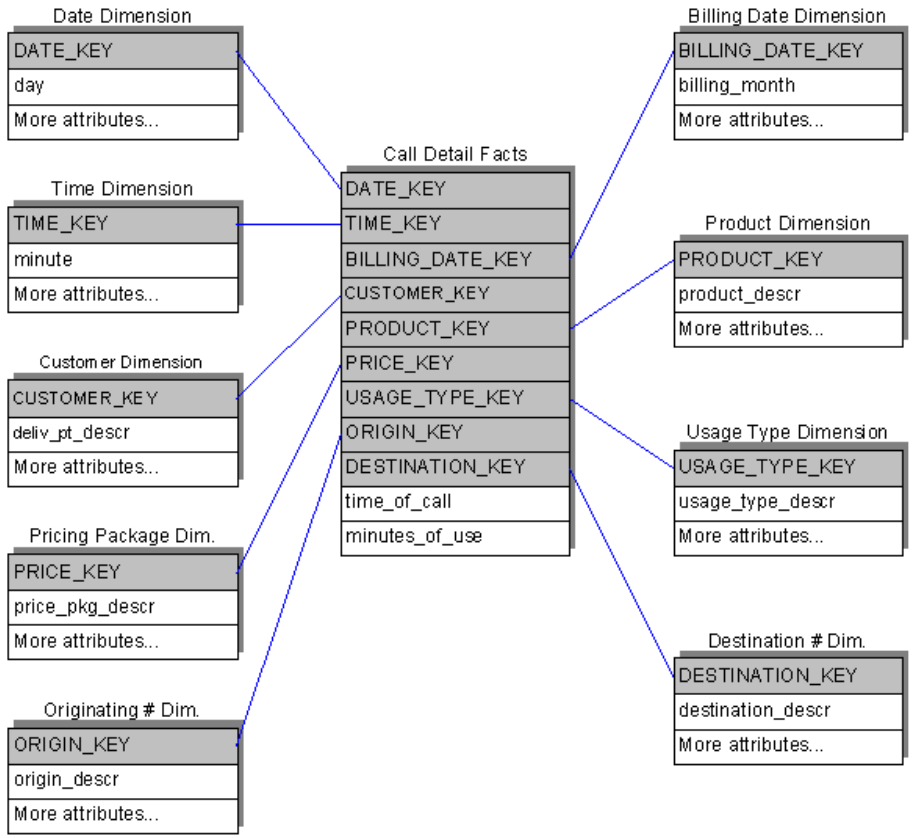
Star Schema ***Call Detail Tracking***



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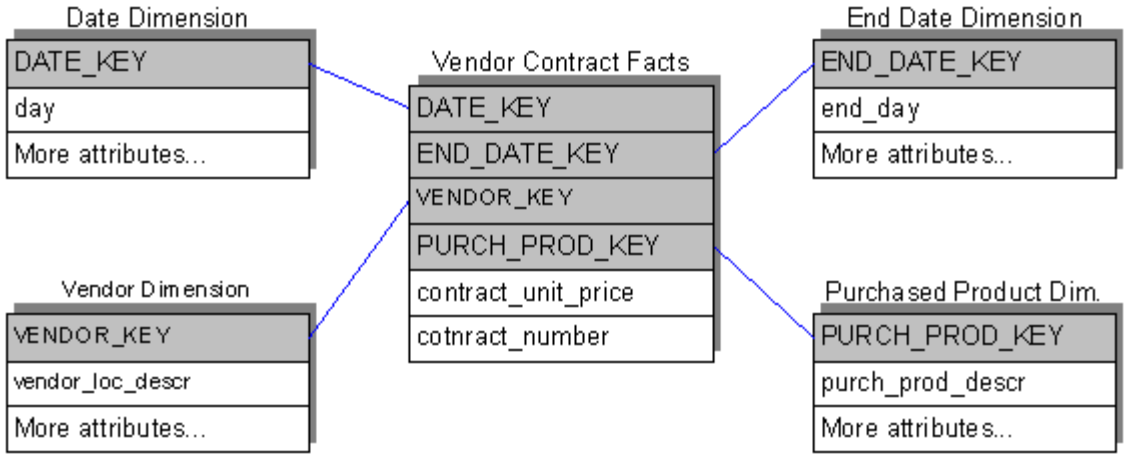
Star Schema *Vendor Contract*



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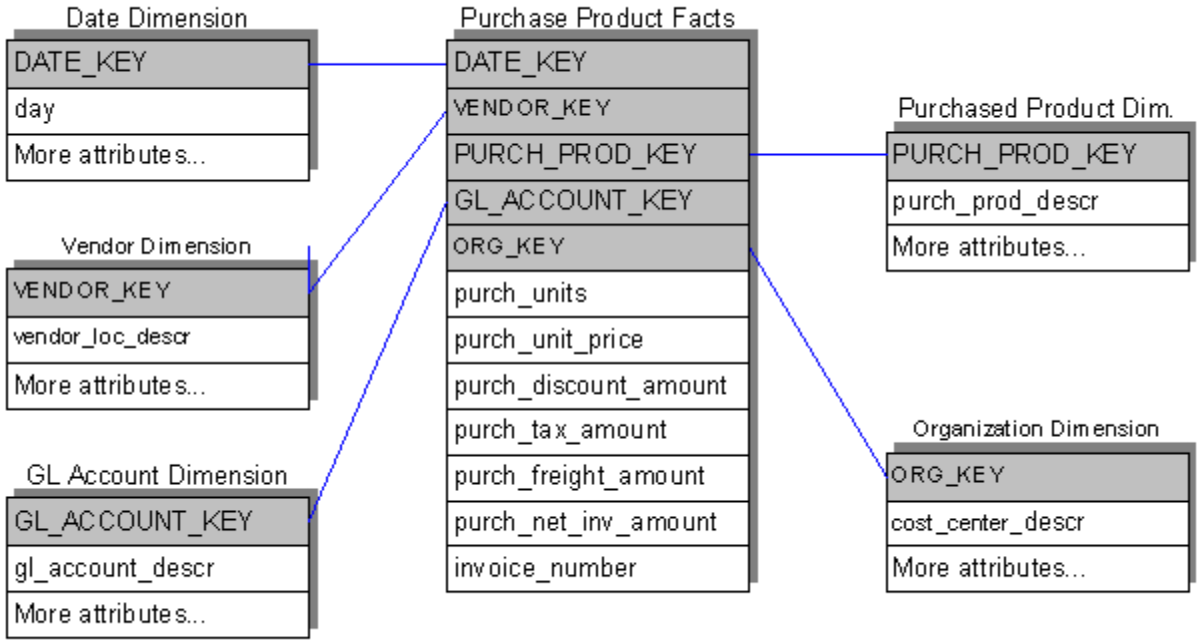
Star Schema *Purchased Product Tracking*



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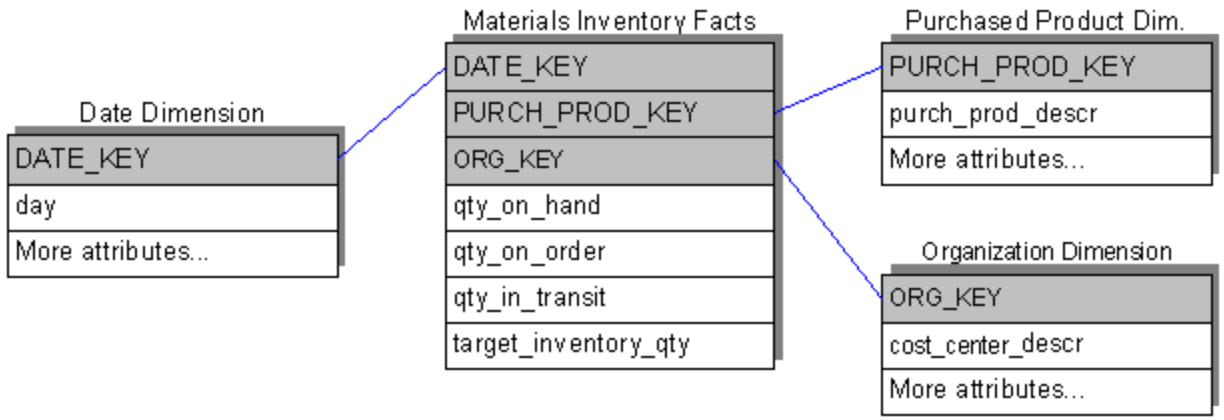
Star Schema *Materials Inventory*



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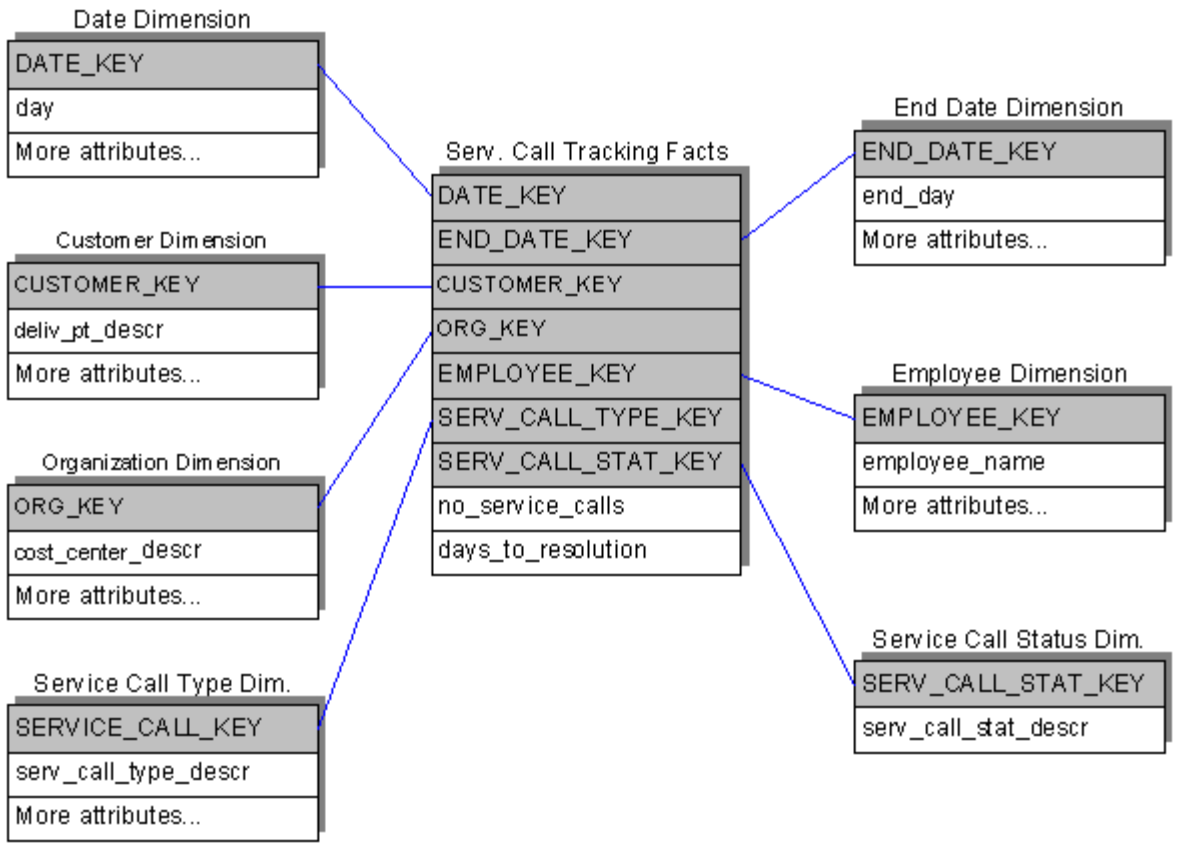
Star Schema *Service Call Tracking*



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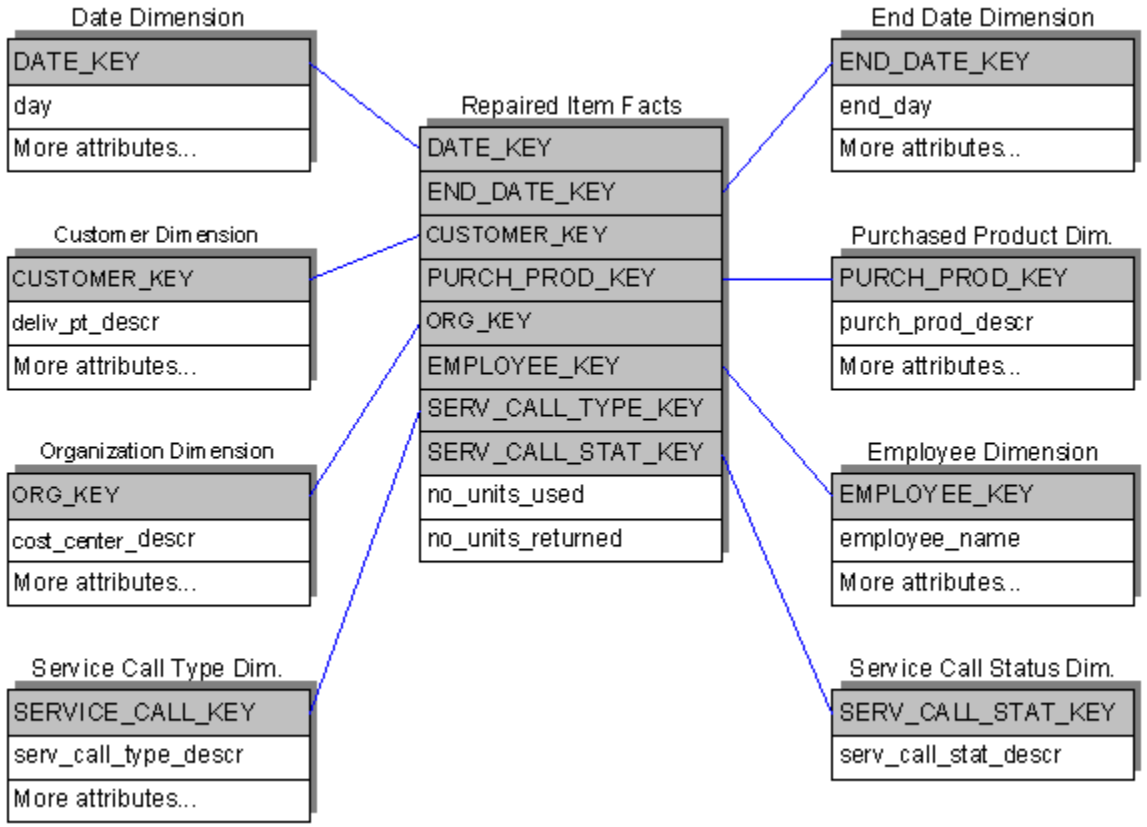
Star Schema *Repaired Items*



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